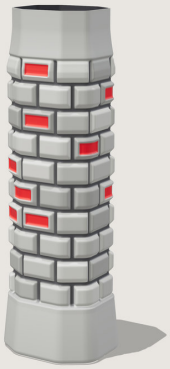
Pickleball Paddle Grips

Ten designs. One paddle slot.

Full specifications, true-size drawings and dimension data for every grip.



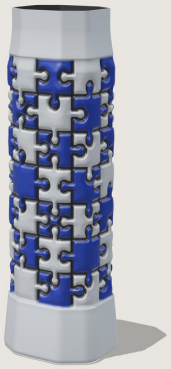
VORTEX GRIP
01 · 57.2 g PLA



CELLULAR MOD
02 · 57.8 g PLA



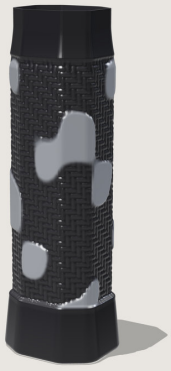
TESSEL-BLOCK
03 · 57.3 g PLA



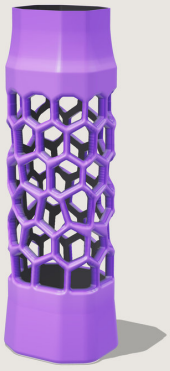
LOGIC-GRIP
04 · 58.9 g PLA



NEURO-TREAD
05 · 57.6 g PLA



CARBON MATRIX
06 · 56.4 g PLA



VORONOI-CORE
07 · 59.5 g PLA



TOPO-FLOW
08 · 57.8 g PLA



ERGO-CONTOUR
09 · 55.5 g PLA



HEXA-MOD
10 · 57.5 g PLA

135.0 mm
overall length, 13.5 cm

30.2 × 16.2 mm
paddle slot, identical on all ten

56 – 59 g
in PLA, sealed hollow core

Ø 3.5 mm
screw hole through the butt

Slot sized to the measured paddle · measured from the STL files · colours are preview only

10 specification sheets · 40 pages

Contents

Click any entry to jump to the page.

FRONT MATTER

All ten at a glance 3
Weight, size and print-readiness compared

The paddle slot and the reference 4
What is identical on every grip, and what changed from the reference sleeve

SPECIFICATION SHEETS · 4 PAGES EACH

01 **Vortex Grip** 5–8
Overview 5 · Drawing 6 · Sections 7 · Surface 8

02 **Cellular Mod** 9–12
Overview 9 · Drawing 10 · Sections 11 · Surface 12

03 **Tessel-Block** 13–16
Overview 13 · Drawing 14 · Sections 15 · Surface 16

04 **Logic-Grip** 17–20
Overview 17 · Drawing 18 · Sections 19 · Surface 20

05 **Neuro-Tread** 21–24
Overview 21 · Drawing 22 · Sections 23 · Surface 24

06 **Carbon Matrix** 25–28
Overview 25 · Drawing 26 · Sections 27 · Surface 28

07 **Voronoi-Core** 29–32
Overview 29 · Drawing 30 · Sections 31 · Surface 32

08 **Topo-Flow** 33–36
Overview 33 · Drawing 34 · Sections 35 · Surface 36

09 **Ergo-Contour** 37–40
Overview 37 · Drawing 38 · Sections 39 · Surface 40

10 **Hexa-Mod** 41–44
Overview 41 · Drawing 42 · Sections 43 · Surface 44

BACK MATTER

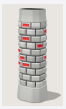
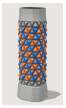
Notes, method and file manifest 45

HOW TO READ THESE SHEETS

Every dimension is measured from the STL, not copied from the design intent. Drawings on the second page of each sheet are true size when printed at 100 %. Renders and colours are a preview; the STLs are single-material. Weight = volume × density at 100 % infill.

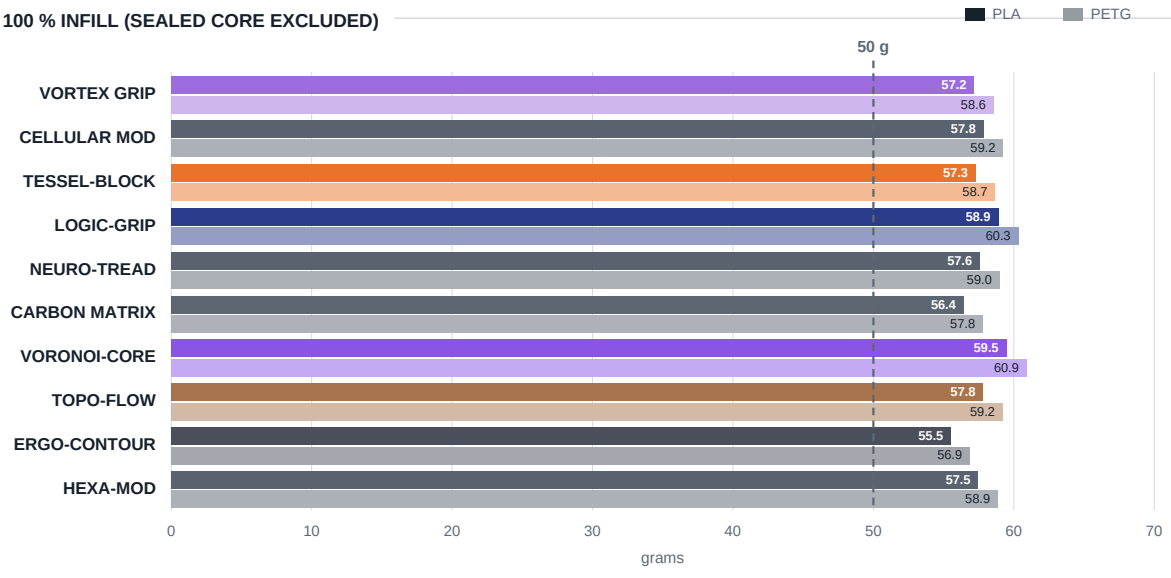
All ten at a glance

Same paddle slot, same foot, same envelope; ten different surfaces, all close to 50 g.

	DESIGN	VOLUME	PLA g	PETG g	TPU g	ABS g	PEAK	BAND W×D	BRIDGE	COLOURS
	**VORTEX GRIP** PG-01	46.1 cm³	**57.2**	58.6	55.8	47.9	2.85	37.8 × 31.7	3.4	
	**CELLULAR MOD** PG-02	46.6 cm³	**57.8**	59.2	56.4	48.5	3.04	38.1 × 31.9	3.4	
	**TESSEL-BLOCK** PG-03	46.2 cm³	**57.3**	58.7	55.9	48.0	3.44	38.9 × 32.9	3.2	
	**LOGIC-GRIP** PG-04	47.5 cm³	**58.9**	60.3	57.5	49.4	2.81	37.7 × 31.7	3.7	
	**NEURO-TREAD** PG-05	46.4 cm³	**57.6**	59.0	56.2	48.3	3.50	39.1 × 33.0	3.4	
	**CARBON MATRIX** PG-06	45.5 cm³	**56.4**	57.8	55.1	47.3	2.73	37.5 × 31.4	3.6	
	**VORONOI-CORE** PG-07	48.0 cm³	**59.5**	60.9	58.0	49.9	3.50	39.1 × 33.0	3.7	
	**TOPO-FLOW** PG-08	46.6 cm³	**57.8**	59.2	56.4	48.5	3.13	38.3 × 32.1	3.4	
	**ERGO-CONTOUR** PG-09	44.8 cm³	**55.5**	56.9	54.2	46.6	2.77	36.9 × 30.9	3.6	
	**HEXA-MOD** PG-10	46.3 cm³	**57.5**	58.9	56.1	48.2	3.50	38.7 × 33.0	3.6	

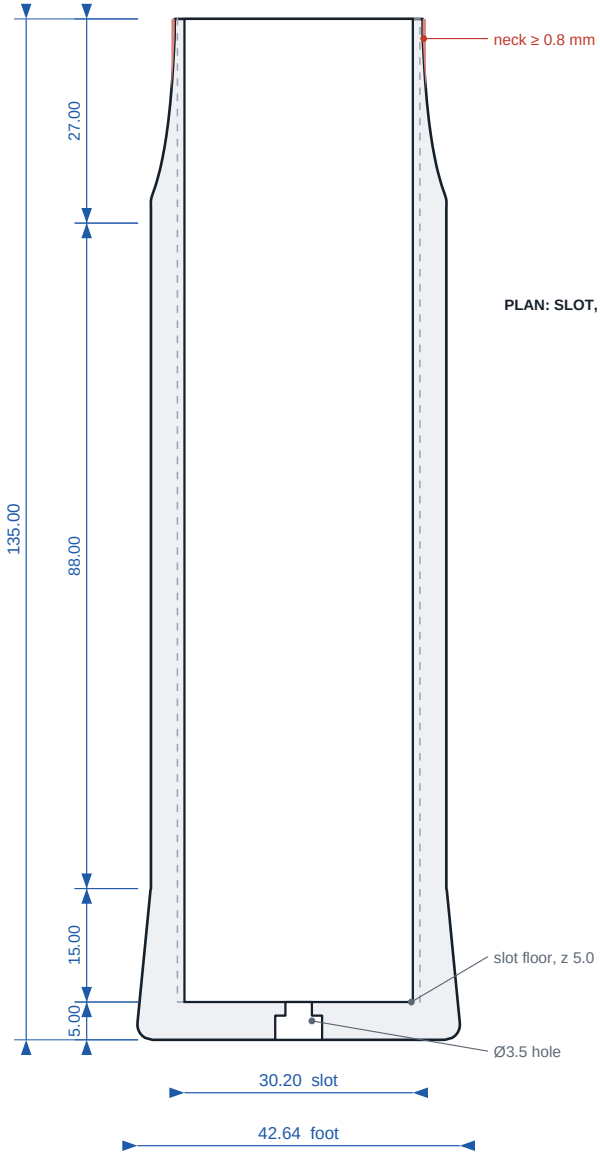
Volume and weights: 100 % infill, volume × density (g/cm³: PLA 1.24, PETG 1.27, TPU 1.21, ABS 1.04). Peak = tallest texture above the base line, mm. Bridge = longest unsupported run, mm.

WEIGHT AT 100 % INFILL (SEALED CORE EXCLUDED)

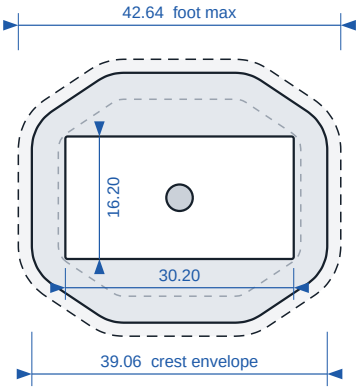


The paddle slot and the reference

Outer envelope measured from PickleballGrip_1.3mf; the interior is a slot for the paddle with a sealed hollow core in the walls.



FRONT ELEVATION OF THE ENVELOPE (1 : 1)



PLAN: SLOT, SCREW HOLE, REFERENCE BORE (DASHED), CREST ENVELOPE, FOOT (1 : 1)

IDENTICAL ON ALL TEN

Paddle slot	**30.2 × 16.2 mm, corner radius 0.3**
Slot depth / floor	**130.00 mm / z 5.0**
Screw hole	**Ø 3.5 mm, z 0 – 5, + Ø 6.2 × 3.2 counterbore**
Overall length / base	**135.0 mm / 5.0 mm**
Foot flare	**z 0 – 20, max 42.63 × 36.61**
Bottom fillet	**about 2 mm**
Crest envelope	**3.5 mm from the reference bore line**
Outer skin	**never exceeds the reference**

CHANGED FROM THE REFERENCE SLEEVE

Interior — **filled; slot 30.2 × 16.2 mm**
Replaces the 32.06 × 26.04 mm bore. Slot = measured paddle tongue 30 × 16 mm + 0.2 mm.

Screw hole — **Ø 3.5 mm + Ø 6.2 × 3.2 counterbore**
An M3 socket-head (cap) screw sits flush with the butt; 1.8 mm of floor is left under its head.

Hollow core — **sealed, 25 – 32 cm³, walls ≈ 1.2 mm**
Only in the thick side walls, z 1 – 110; the head above z 110 is solid. Roofs ≤ 45°, no supports.

Length — **135.0 mm = 5.0 base + 130.0 slot**
Rev 4 was 132.8 mm: the paddle neck flare rested on the rim and pried it open. 0.5 mm lead-in chamfer at the mouth.

Weight — **56 – 59 g in PLA**
Rev 4 with the hollow head was 50 – 54 g (it cracked); rev 3 fully solid 88 – 98 g.

ENVELOPE VS HEIGHT: REFERENCE AND GRIPS



HOW IT WAS VERIFIED

- Slot: the cavity around the axis from z 5 to the top equals the slot outline; no material inside it (0.000 mm²).
- Screw hole: one round Ø 3.5 mm hole on the axis, Ø 6.2 × 3.2 counterbore at the butt face, floor solid around it.
- Envelope: outer skin vs the reference, worst case +0.003 mm.
- Hollow core: wall to the nearest outer surface measured in 3D (median 1.2 mm, min 1.1 mm); no cavity above z 110.
- Print check: every 0.20 mm layer sliced; one outer body, no material under 0.4 mm wide, flat bridges ≤ 4 mm.

01 VORTEX GRIP

Braided over/under helical straps for dynamic multi-point contact.

57.2 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE

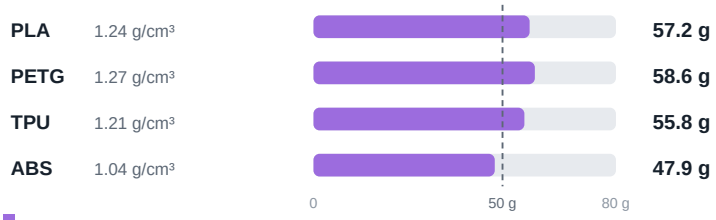


3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**37.8 × 31.7 mm**
Peak texture height	**2.85 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 3.2 counterbore**
Material volume / surface	**46.10 cm³ / 460 cm²**

WEIGHT · 100 % INFILL



COLOURWAY



#3A225C #9C6CDE

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**25.5 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 19.2 m (100 % flow)**

DESIGN INTENT

Sixteen straps wound in opposite directions cross in an over/under braid, leaving dark valleys between them. Every crossing gives the hand an edge to bite on, so the grip works from any hand position on the handle.

Straps rise no faster than 45 degrees, so no supports are needed.

KEY FEATURES

- 8 right-hand + 8 left-hand straps on a 36 degree helix
- Over/under weave: the strap on top alternates at every crossing
- Valleys drop to about the 1.2 mm base wall, which keeps the mass low

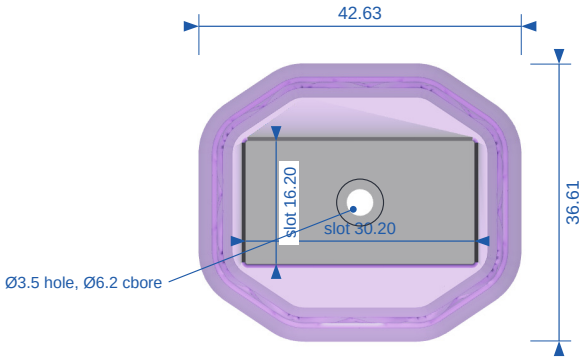
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 3.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 1**
Wall around the cavity (min)	**1.15 mm**
Material under 0.4 mm wide	**0.000 % of volume**
Widest flat bridge	**3.4 mm**

FILES

stl/01_vortex_grip.stl	19.9 MB · 398,552 triangles
3mf/01_vortex_grip.3mf	4.2 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

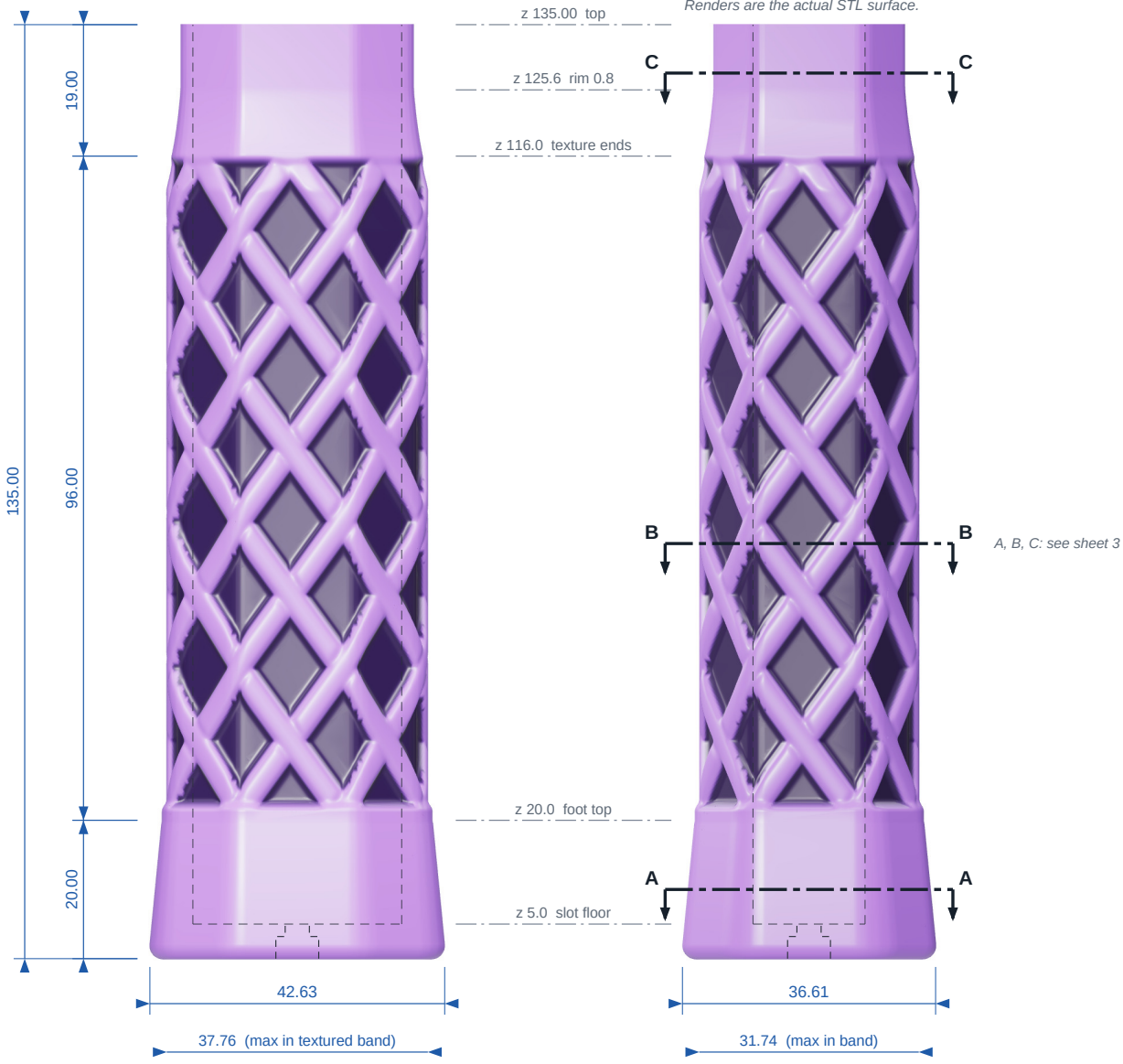


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**37.76 × 31.74**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**2.85**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 3.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1

0 10 20 30 40 50 mm

PART

VORTEX GRIP

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

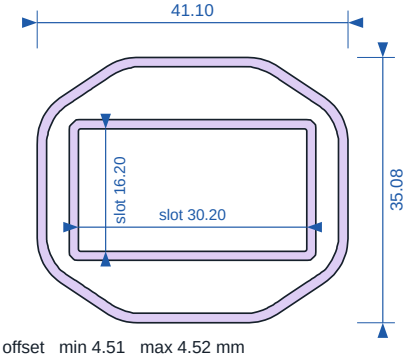
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 110 is solid.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **57.2 g**	VOLUME **46.10 cm³**	SURFACE AREA **460 cm²**
REVISION **5**	DATE **2026-10-19**	SHEET **2 of 4**

Sections & profiles

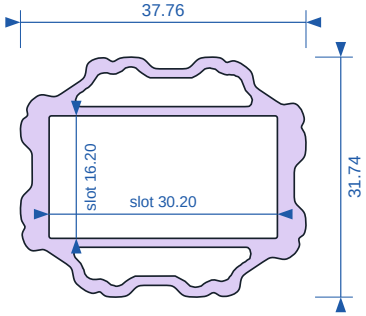
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



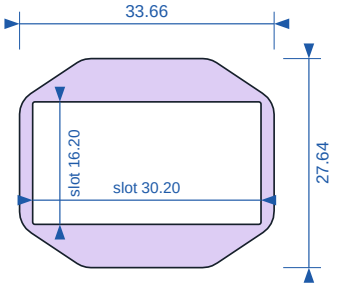
Skin offset min 4.51 max 4.52 mm
Area 278 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



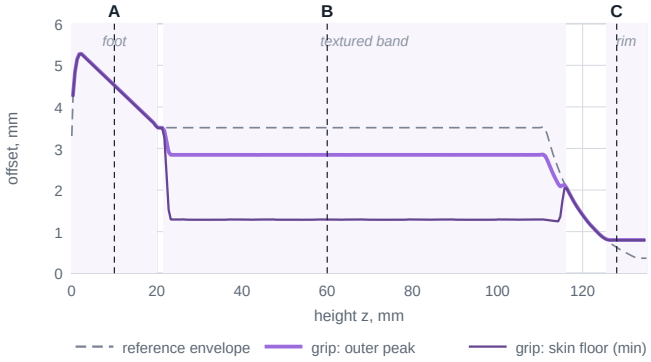
Skin offset min 1.29 max 2.85 mm
Area 307 mm² · outer perimeter 118.0 mm

SECTION C–C z = 128 mm · rim

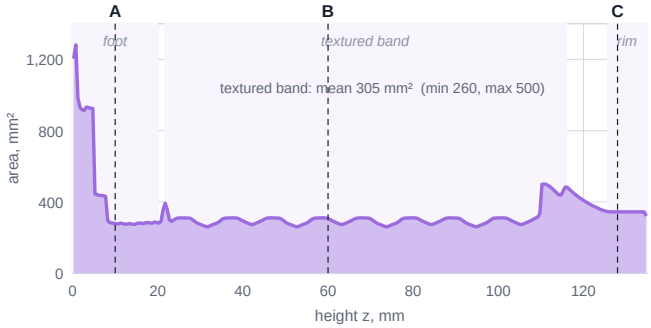


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 3.2**
Sealed cavity / min wall	**25.5 cm³ / 1.15 mm**
Outer W × D in band (max)	**37.76 × 31.74 mm**
Material volume / surface	**46.10 cm³ / 460.1 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	135.00	**9.4**

Relief floor above reference line (design / min)	**1.20 / 1.29 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**2.85 mm**

OUTER DIMENSIONS BY HEIGHT

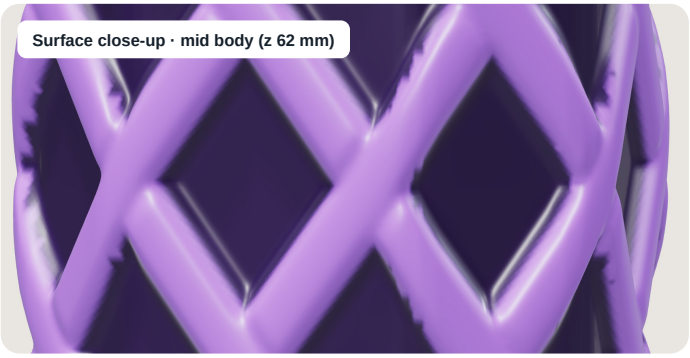
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	684	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	283	3.52	3.52
30.0	37.76	31.74	114.5	274	1.29	2.85
45.0	37.76	31.74	118.0	302	1.29	2.85
60.0	37.76	31.74	118.0	307	1.29	2.85
75.0	37.76	31.74	114.1	272	1.29	2.85
90.0	37.76	31.74	117.1	310	1.29	2.85
105.0	37.75	31.74	117.1	275	1.29	2.85
116.0	36.24	30.22	113.4	483	2.04	2.09
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.80	0.80

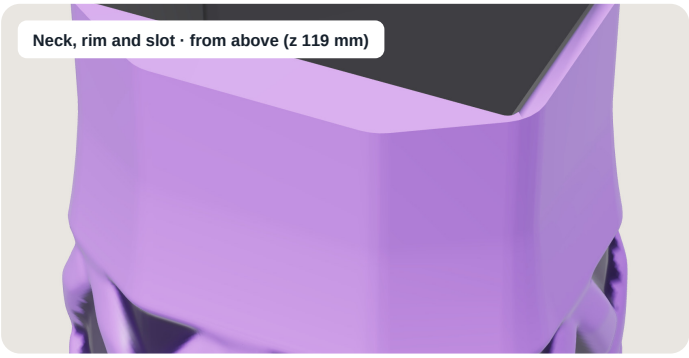
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

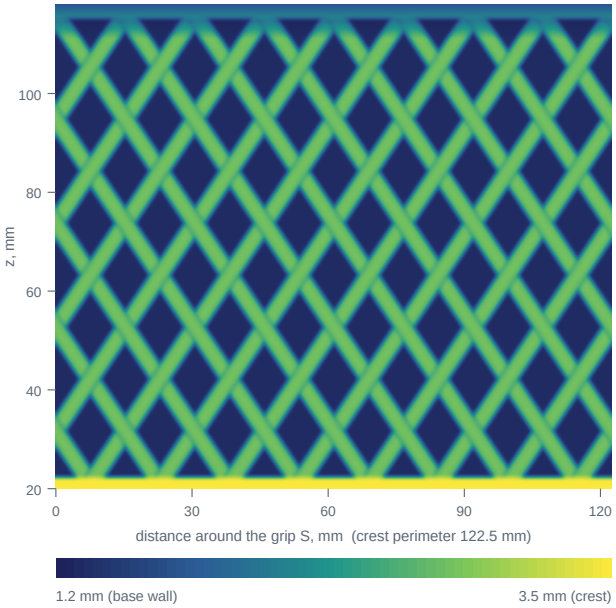


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Straps	**8 + 8, counter-wound**
Helix angle	**36 deg from vertical**
Strap pitch (around)	**15.31 mm**
Strap width	**6.1 mm around / 5.0 mm across**
Strap top height	**2.35 – 2.86 mm above base**
Valley floor	**1.29 mm above base**
Over/under step	**0.37 mm**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 1**
Wall around the cavity (min)	**1.15 mm**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Widest flat bridge	**3.4 mm**

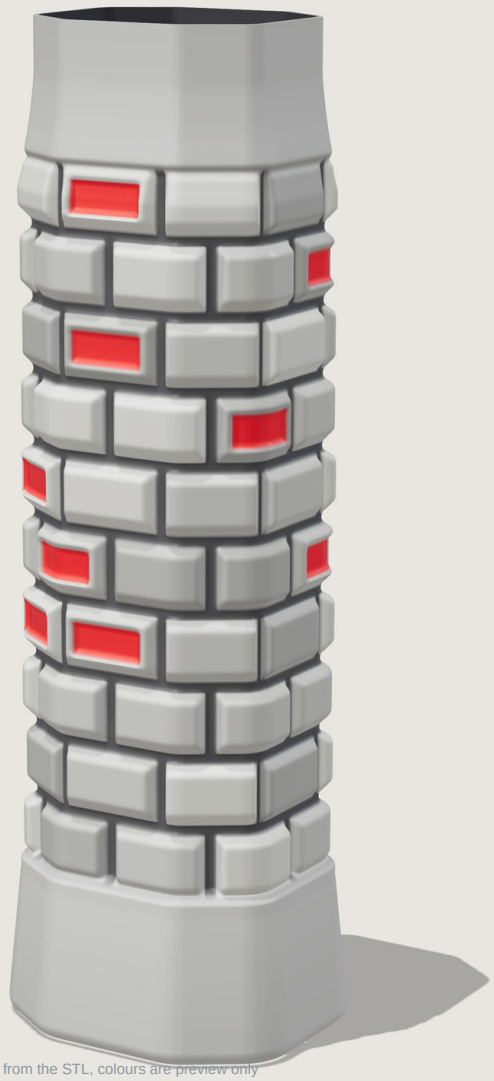
02 CELLULAR MOD

Staggered stacked cells; red cells are recessed pockets for swappable inserts.

57.8 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE

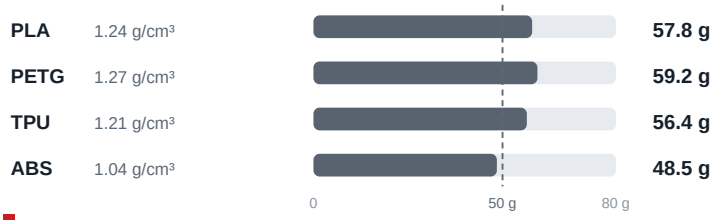


3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**38.1 × 31.9 mm**
Peak texture height	**3.04 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 3.2 counterbore**
Material volume / surface	**46.64 cm³ / 465 cm²**

WEIGHT · 100 % INFILL



COLOURWAY



#464648 #969692 #C81E22

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**25.9 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 19.4 m (100 % flow)**

DESIGN INTENT

A wall of staggered, individually pillowed cells. Roughly one cell in five is recessed as an insert pocket (red in the colourway), sized to take a contrasting insert. No insert parts are included.

Cell edges are chamfered; nothing overhangs beyond 45 degrees.

KEY FEATURES

- 10 rows × 8 staggered cells with 1.4 mm gaps
- Three cell heights give a subtly stepped surface
- About 22% of cells are recessed 0.7 – 1.2 mm as insert pockets

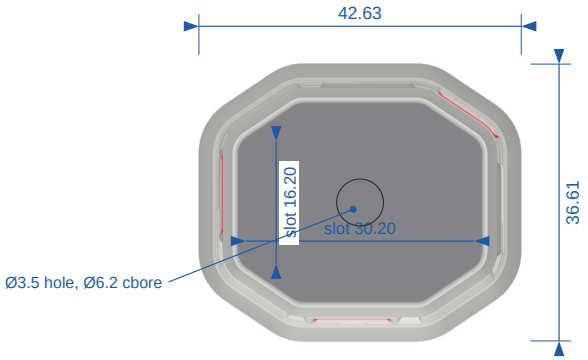
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 3.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 1**
Wall around the cavity (min)	**1.10 mm**
Material under 0.4 mm wide	**0.000 % of volume**
Widest flat bridge	**3.4 mm**

FILES

stl/02_cellular_mod.stl	9.9 MB · 198,210 triangles
3mf/02_cellular_mod.3mf	2.1 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

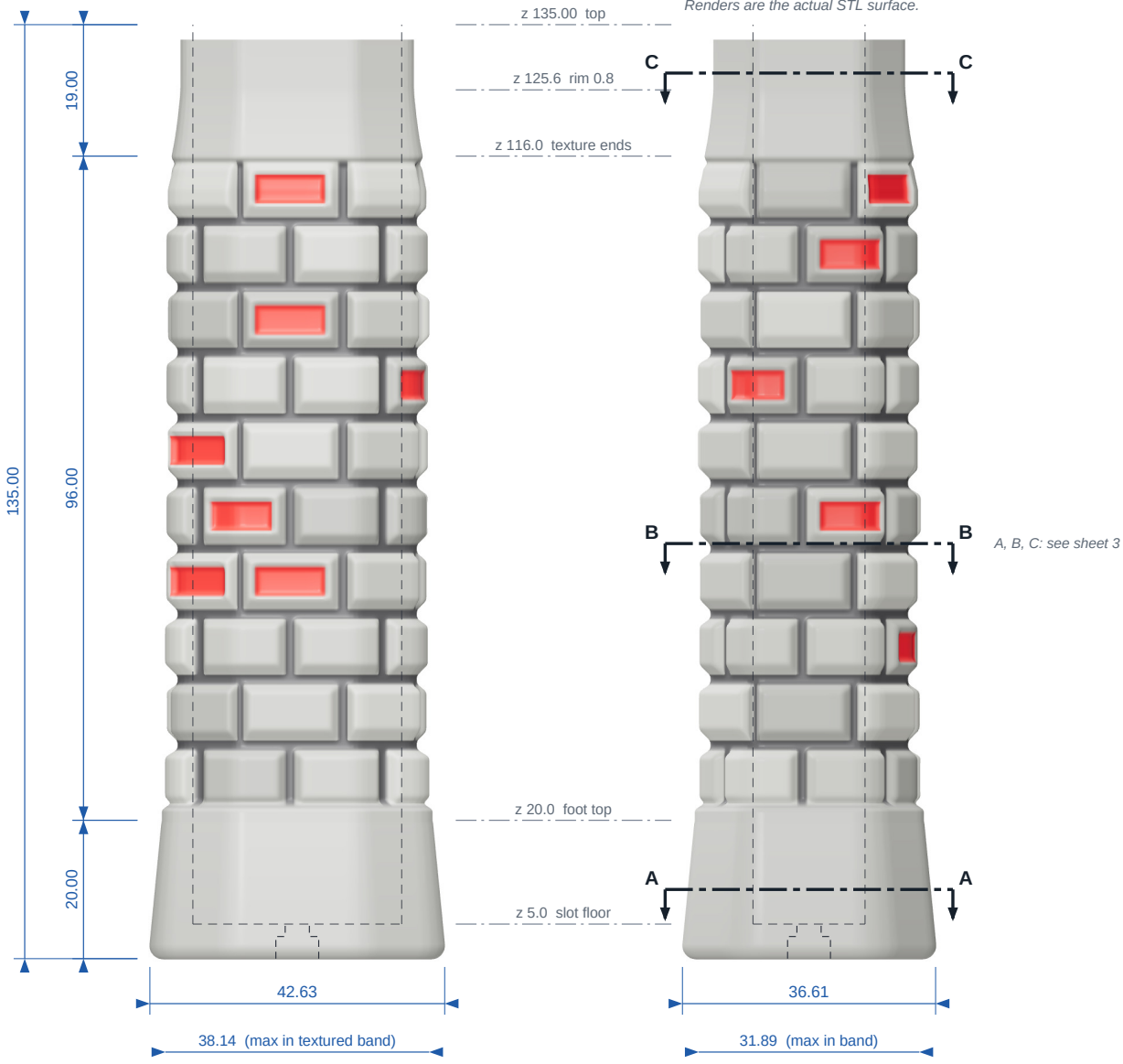


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**38.14 × 31.89**
Foot top (z 20)	**39.15 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.04**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 3.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

CELLULAR MOD

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

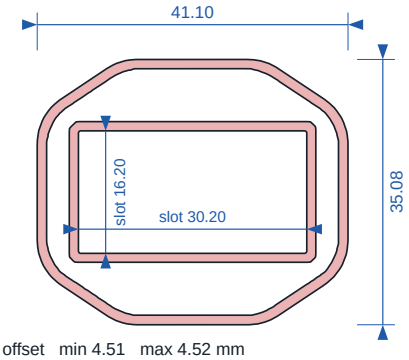
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 110 is solid.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE	UNITS	PROJECTION
1 : 1	millimetres	
MATERIAL	**INFILL**	**LAYER HEIGHT**
PLA / PETG / TPU	100 %	0.20 mm
MASS (PLA)	**VOLUME**	**SURFACE AREA**
57.8 g	46.64 cm³	465 cm²
REVISION	**DATE**	**SHEET**
5	2026-10-19	2 of 4

Sections & profiles

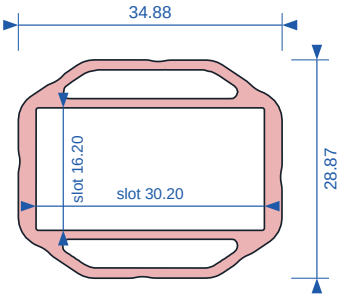
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



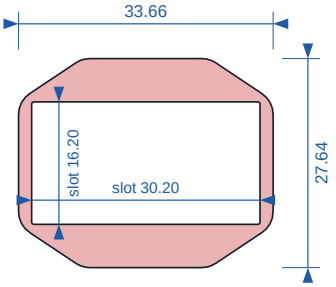
Skin offset min 4.51 max 4.52 mm
Area 278 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



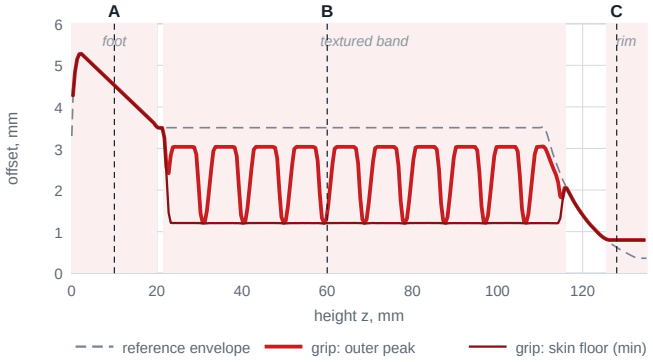
Skin offset min 1.20 max 1.46 mm
Area 249 mm² · outer perimeter 109.3 mm

SECTION C–C z = 128 mm · rim

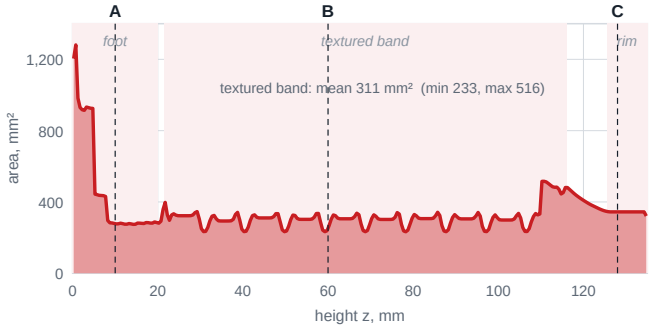


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 3.2**
Sealed cavity / min wall	**25.9 cm³ / 1.10 mm**
Outer W × D in band (max)	**38.14 × 31.89 mm**
Material volume / surface	**46.64 cm³ / 464.9 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	135.00	**9.4**

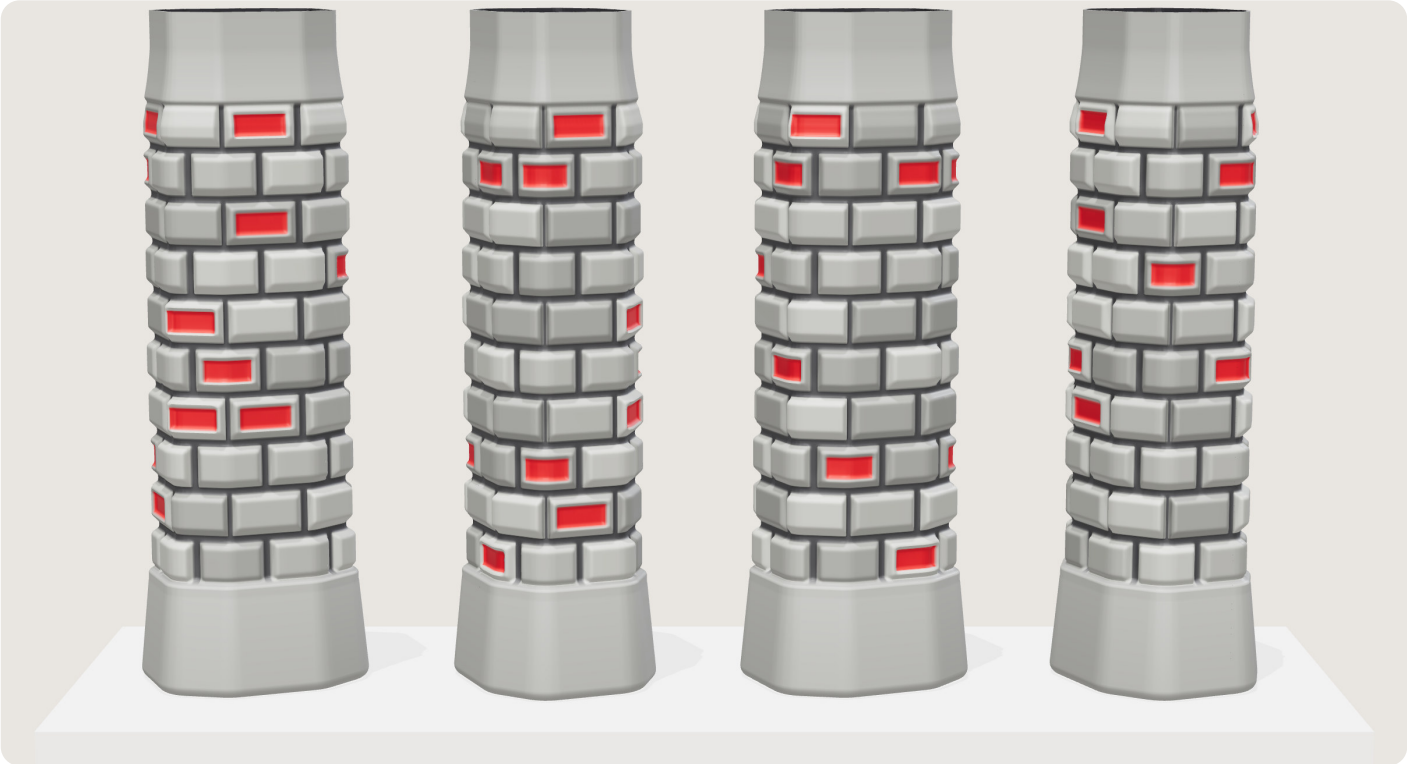
Relief floor above reference line (design / min)	**1.20 / 1.20 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.04 mm**

OUTER DIMENSIONS BY HEIGHT

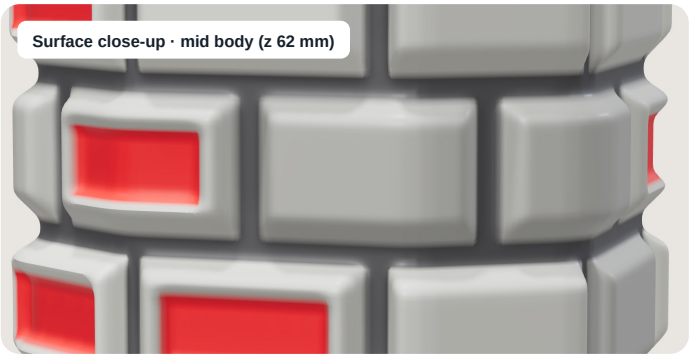
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	684	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	283	3.52	3.52
30.0	35.66	29.57	111.5	277	1.21	1.80
45.0	37.91	31.43	117.0	310	1.21	3.04
60.0	34.96	28.95	109.6	253	1.20	1.46
75.0	37.46	31.66	117.7	303	1.21	3.04
90.0	37.54	31.46	117.3	323	1.21	2.91
105.0	36.98	30.96	115.5	323	1.21	2.64
116.0	36.16	30.14	113.4	481	2.04	2.05
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.80	0.80

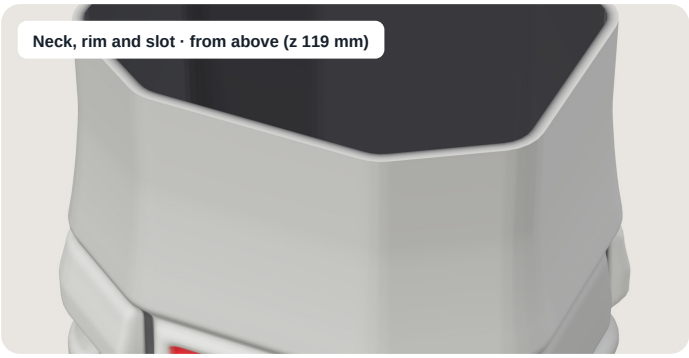
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

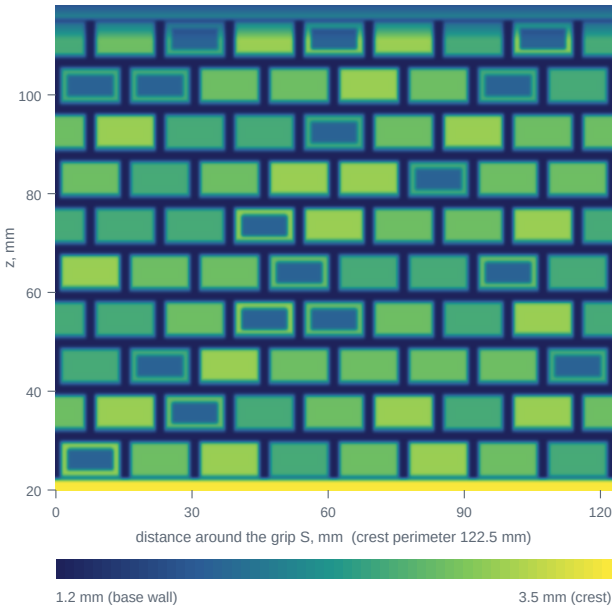


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Rows × cells	**10 × 8 (staggered)**
Row pitch	**9.45 mm**
Cell size	**15.31 × 8.05 mm**
Gap between cells	**1.4 mm**
Cell heights	**2.58 / 2.81 / 3.04 mm**
Pocket floor	**1.84 mm above base**
Groove floor	**1.20 mm (base wall)**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 1**
Wall around the cavity (min)	**1.10 mm**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Widest flat bridge	**3.4 mm**

03 TESSEL-BLOCK

Faceted triangular pyramids in staggered rows for enhanced traction.

57.3 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE

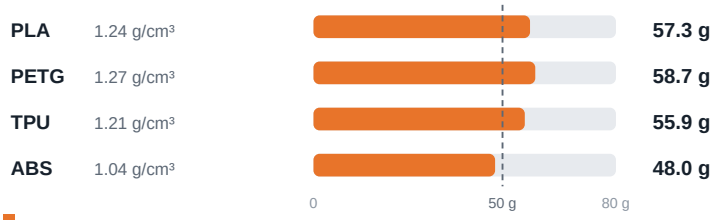


3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**38.9 × 32.9 mm**
Peak texture height	**3.44 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 3.2 counterbore**
Material volume / surface	**46.19 cm³ / 462 cm²**

WEIGHT · 100 % INFILL



COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**25.4 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 19.2 m (100 % flow)**

DESIGN INTENT

A tessellation of triangular pyramids. Traction comes from many small angled edges rather than one deep texture, and up-pointing pyramids stand taller than down-pointing ones, giving a two-level, gem-like surface.

Facet slopes stay below 45 degrees; rim rows are flat grey bands in the colourway.

KEY FEATURES

- 11 rows × 24 triangles (12 up + 12 down per row)
- Sharp V-grooves between facets
- Up-pyramids 3.44 mm, down-pyramids 3.04 mm above the base

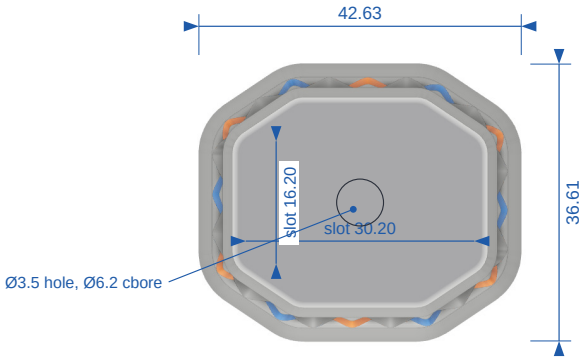
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 3.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 1**
Wall around the cavity (min)	**1.10 mm**
Material under 0.4 mm wide	**0.000 % of volume**
Widest flat bridge	**3.2 mm**

FILES

stl/03_tessel_block.stl	20.9 MB · 417,132 triangles
3mf/03_tessel_block.3mf	4.5 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

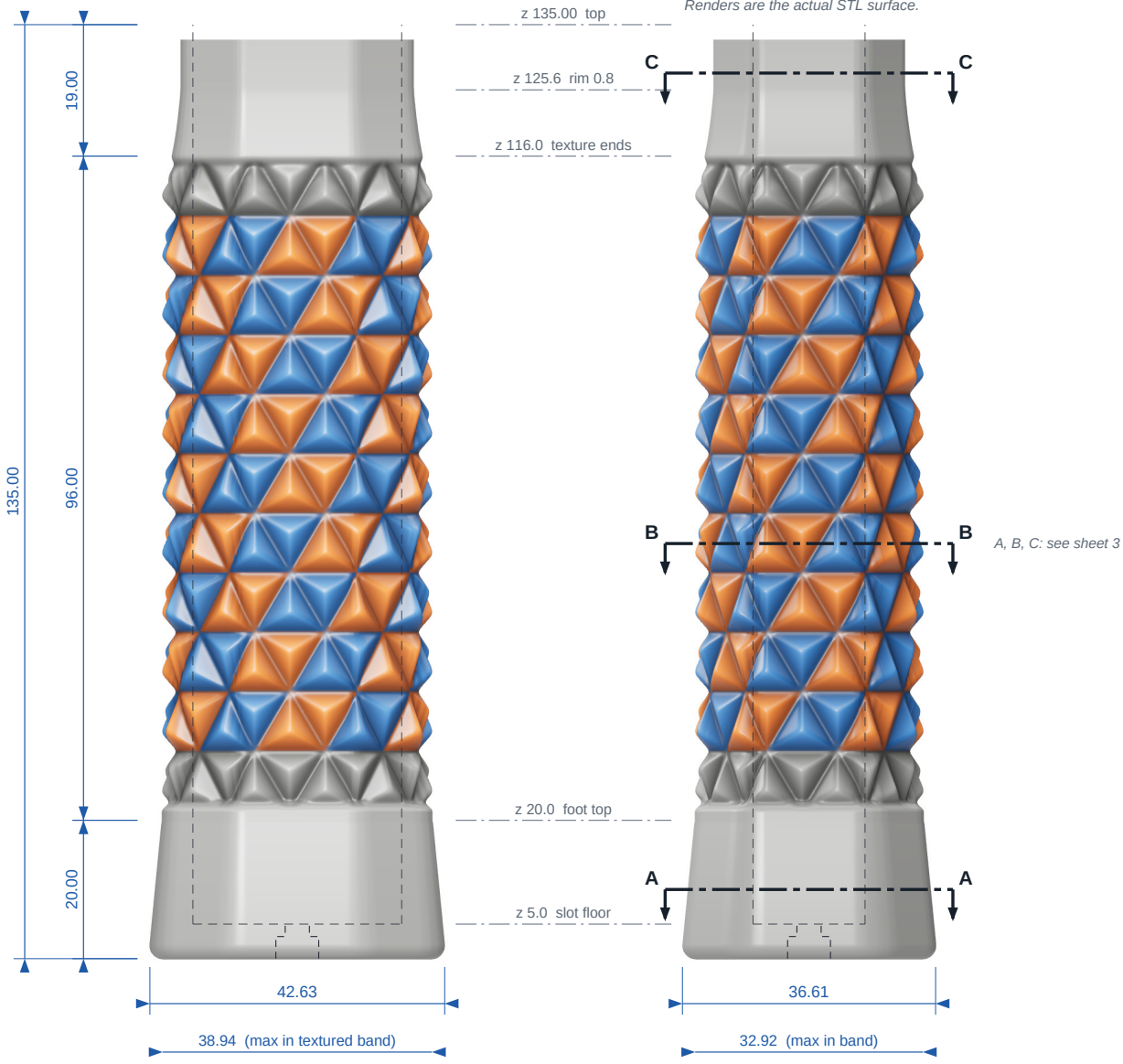


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**38.94 × 32.92**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.44**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 3.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

TESSEL-BLOCK

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

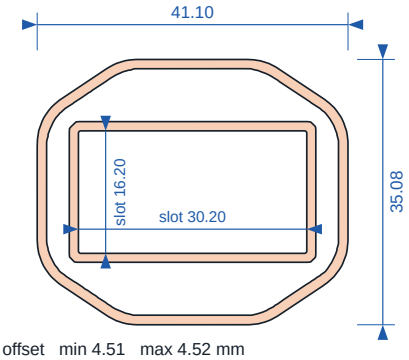
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 110 is solid.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE	UNITS	PROJECTION
1 : 1	millimetres	
MATERIAL	**INFILL**	**LAYER HEIGHT**
PLA / PETG / TPU	100 %	0.20 mm
MASS (PLA)	**VOLUME**	**SURFACE AREA**
57.3 g	46.19 cm³	462 cm²
REVISION	**DATE**	**SHEET**
5	2026-10-19	2 of 4

Sections & profiles

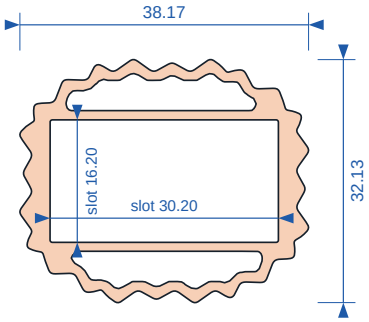
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



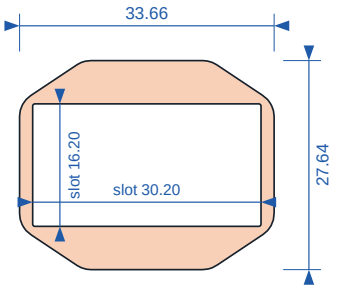
Skin offset min 4.51 max 4.52 mm
Area 278 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



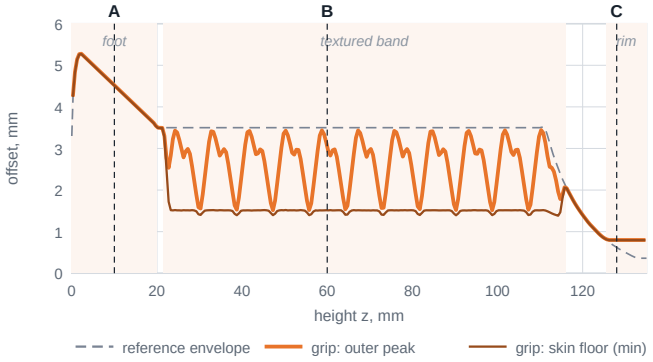
Skin offset min 1.51 max 3.04 mm
Area 305 mm² · outer perimeter 117.2 mm

SECTION C–C z = 128 mm · rim

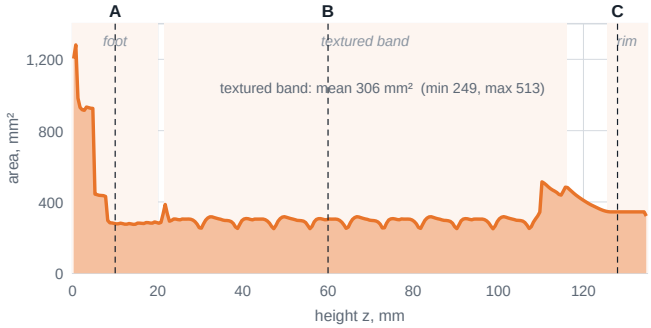


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 3.2**
Sealed cavity / min wall	**25.4 cm³ / 1.10 mm**
Outer W × D in band (max)	**38.94 × 32.92 mm**
Material volume / surface	**46.19 cm³ / 462.3 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	135.00	**9.4**

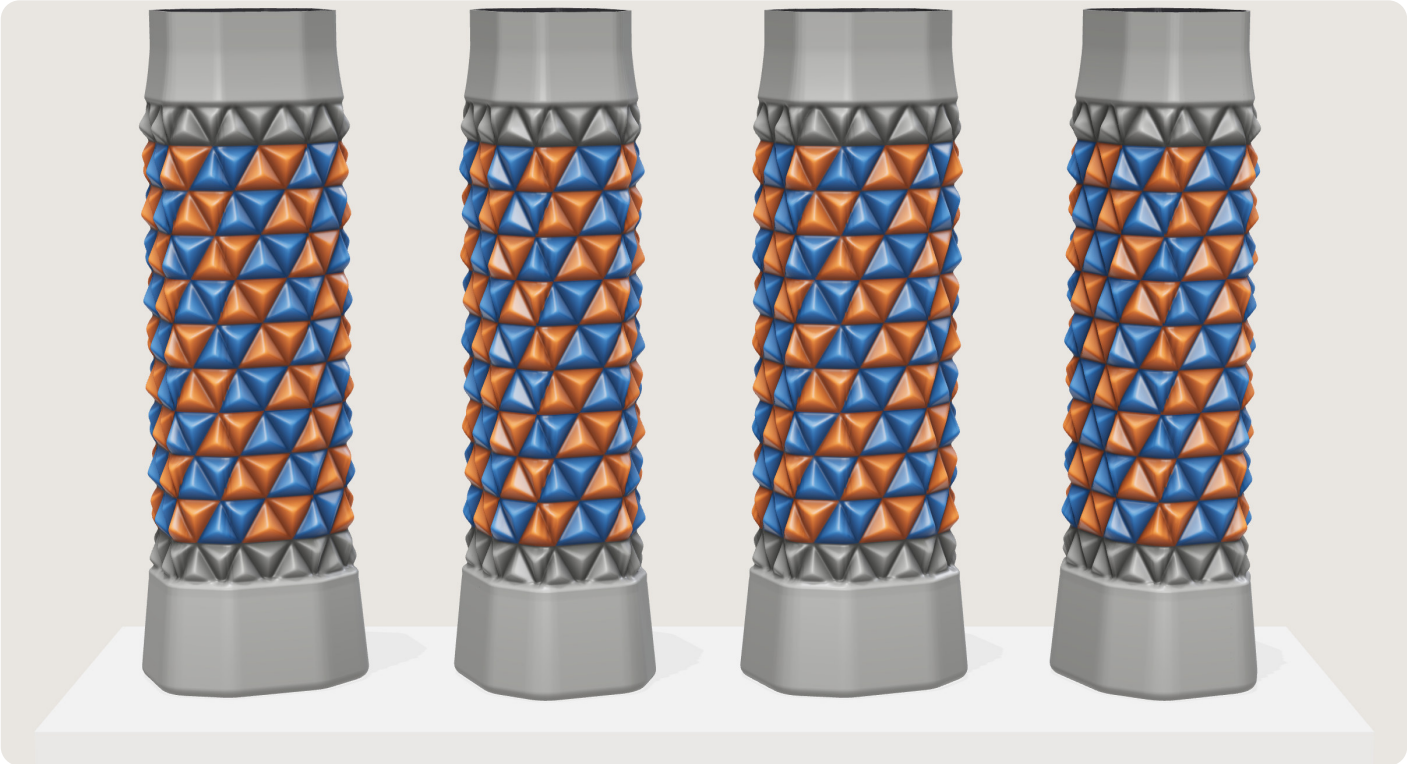
Relief floor above reference line (design / min)	**1.20 / 1.39 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.44 mm**

OUTER DIMENSIONS BY HEIGHT

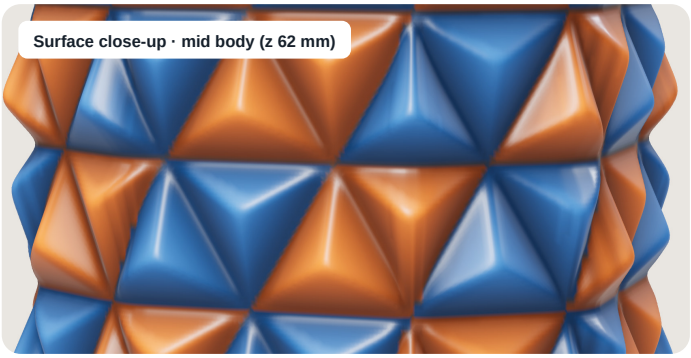
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	684	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	283	3.52	3.52
30.0	35.21	29.18	110.3	254	1.41	1.57
45.0	37.70	31.68	117.3	302	1.52	2.82
60.0	38.17	32.13	117.3	303	1.51	3.04
75.0	38.05	32.03	116.9	306	1.51	3.00
90.0	35.24	29.22	110.3	256	1.42	1.59
105.0	37.81	31.79	115.9	296	1.51	2.87
116.0	36.18	30.16	113.4	482	2.05	2.06
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.80	0.80

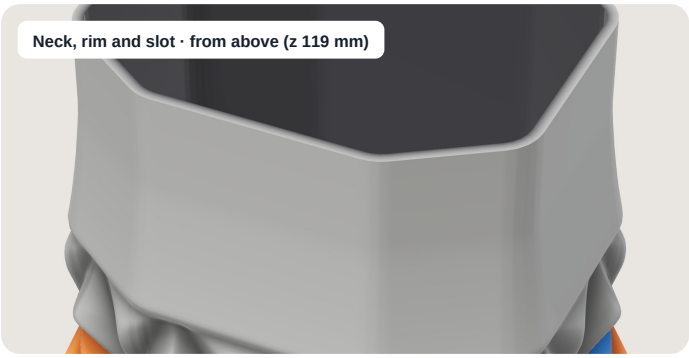
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

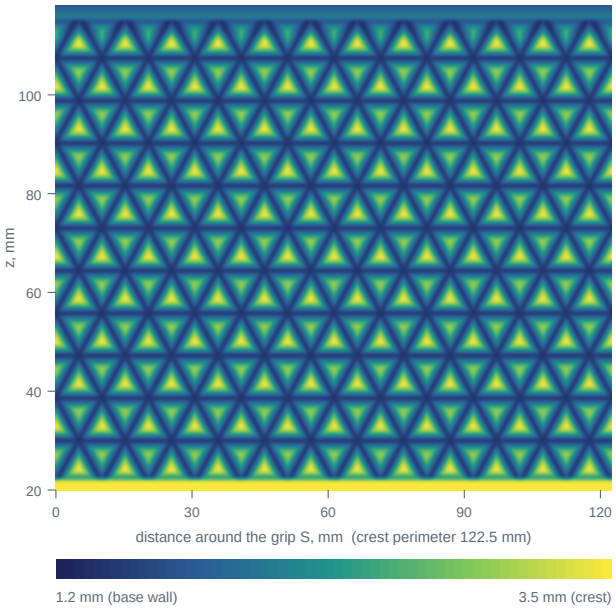


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Rows	**11**
Triangles per row	**24 (12 up + 12 down)**
Triangle base	**10.21 mm**
Row height	**8.59 mm**
Up-pyramid apex	**3.44 mm above base**
Down-pyramid apex	**3.04 mm above base**
Groove floor	**about 1.4 mm above base**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 1**
Wall around the cavity (min)	**1.10 mm**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Widest flat bridge	**3.2 mm**

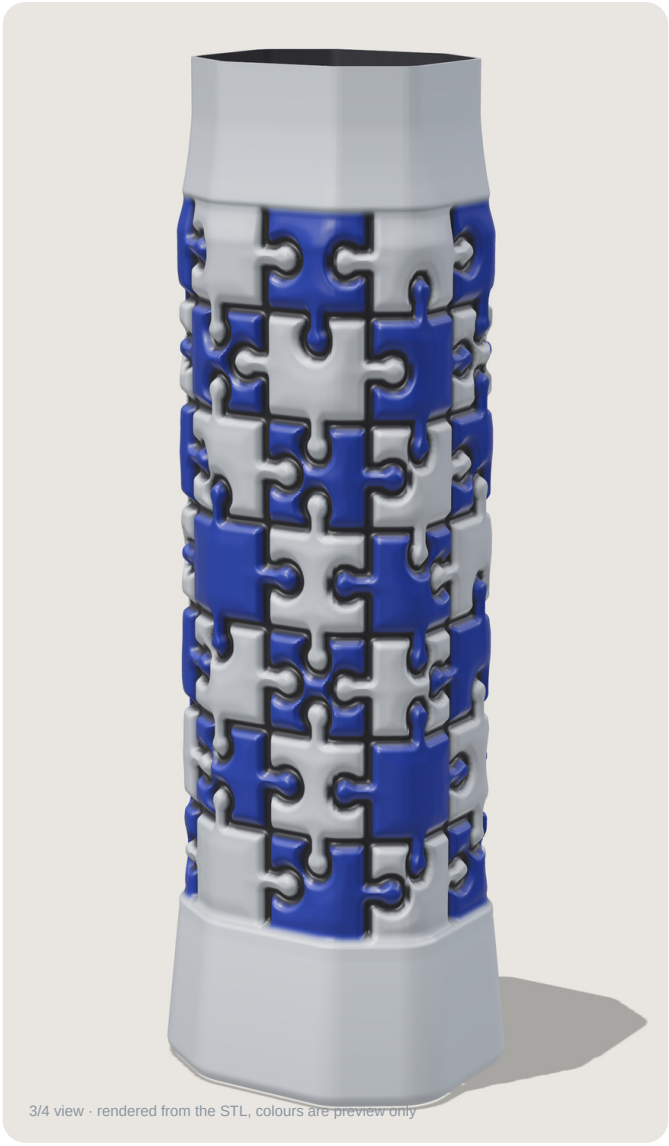
04 LOGIC-GRIP

Seamlessly interlocking ergonomic puzzle design.

58.9 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE

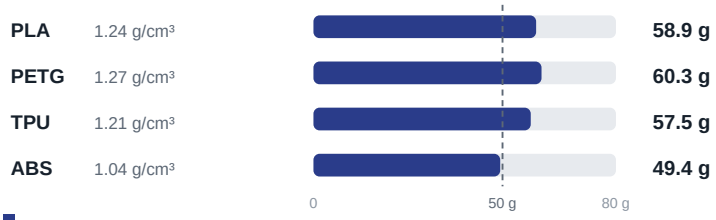


3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**37.7 × 31.7 mm**
Peak texture height	**2.81 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 3.2 counterbore**
Material volume / surface	**47.49 cm³ / 470 cm²**

WEIGHT · 100 % INFILL



COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**25.3 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 19.7 m (100 % flow)**

DESIGN INTENT

Fifty-six jigsaw pieces tile the grip in alternating colours. Each piece is a shallow pillow separated by fine grooves and locked to its neighbours by mushroom knobs; knob directions are randomised so no two rows repeat.

Grooves are 0.84 mm wide, about two nozzle widths, and 1.6 mm deep.

KEY FEATURES

- 8 × 7 = 56 pieces in a checkerboard colourway
- Mushroom knobs: 2.9 mm neck, 4.2 mm head, 5.1 mm reach
- 0.84 mm grooves cut down to the base wall

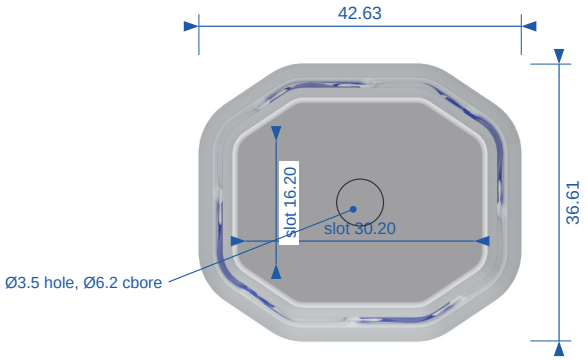
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 3.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 1**
Wall around the cavity (min)	**1.10 mm**
Material under 0.4 mm wide	**0.000 % of volume**
Widest flat bridge	**3.7 mm**

FILES

stl/04_logic_grip.stl	19.6 MB · 391,124 triangles
3mf/04_logic_grip.3mf	4.1 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

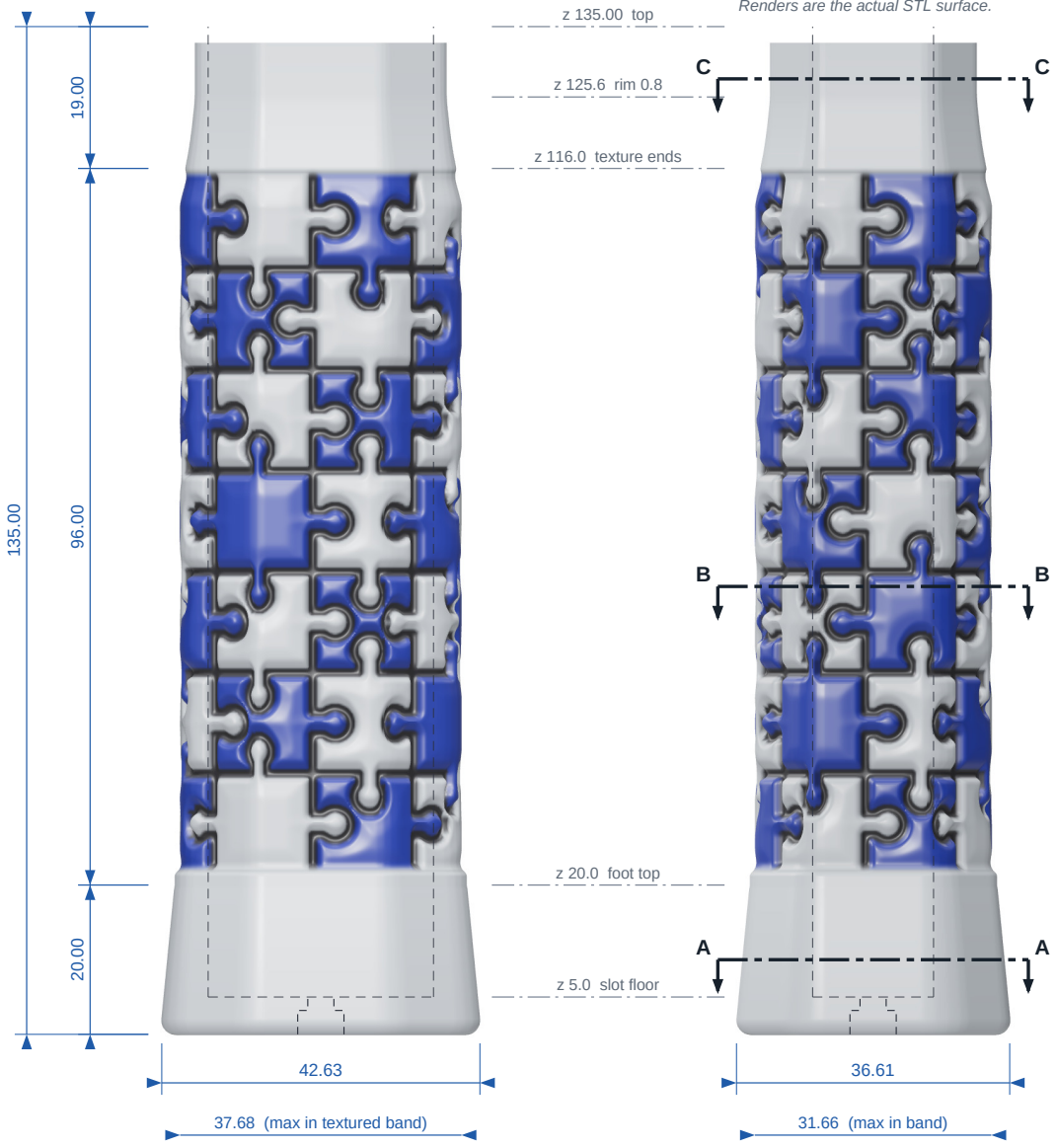


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**37.68 × 31.66**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**2.81**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 3.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

LOGIC-GRIP

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

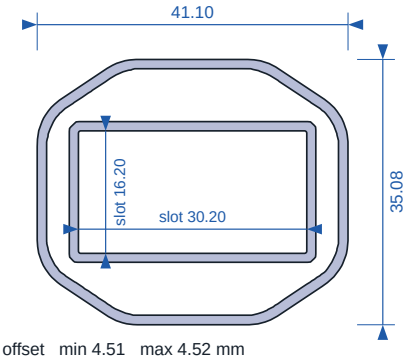
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 110 is solid.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE	UNITS	PROJECTION
1 : 1	millimetres	
MATERIAL	**INFILL**	**LAYER HEIGHT**
PLA / PETG / TPU	100 %	0.20 mm
MASS (PLA)	**VOLUME**	**SURFACE AREA**
58.9 g	47.49 cm³	470 cm²
REVISION	**DATE**	**SHEET**
5	2026-10-19	2 of 4

Sections & profiles

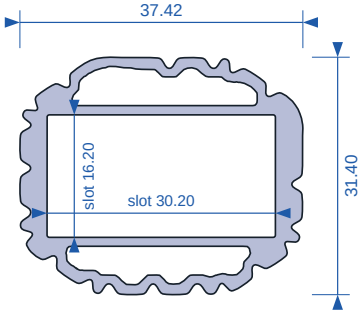
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



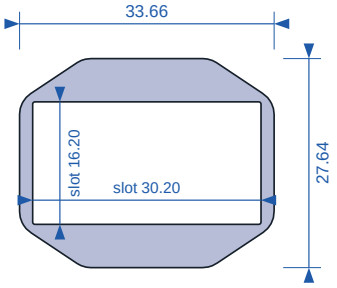
Skin offset min 4.51 max 4.52 mm
Area 278 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



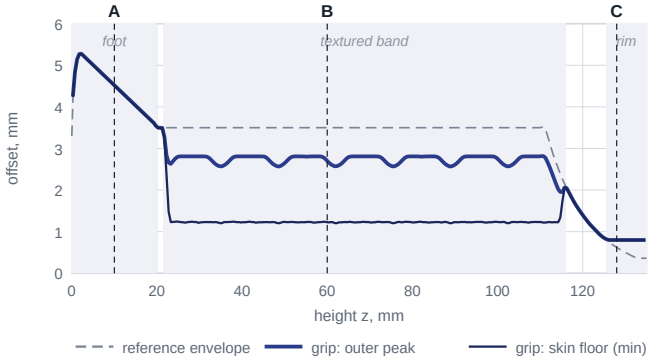
Skin offset min 1.23 max 2.70 mm
Area 315 mm² · outer perimeter 116.7 mm

SECTION C–C z = 128 mm · rim

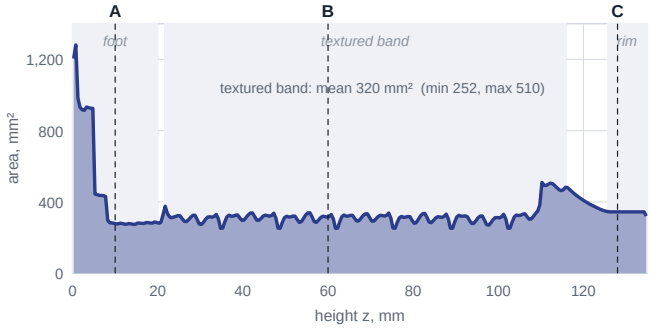


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 3.2**
Sealed cavity / min wall	**25.3 cm³ / 1.10 mm**
Outer W × D in band (max)	**37.68 × 31.66 mm**
Material volume / surface	**47.49 cm³ / 469.5 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	135.00	**9.4**

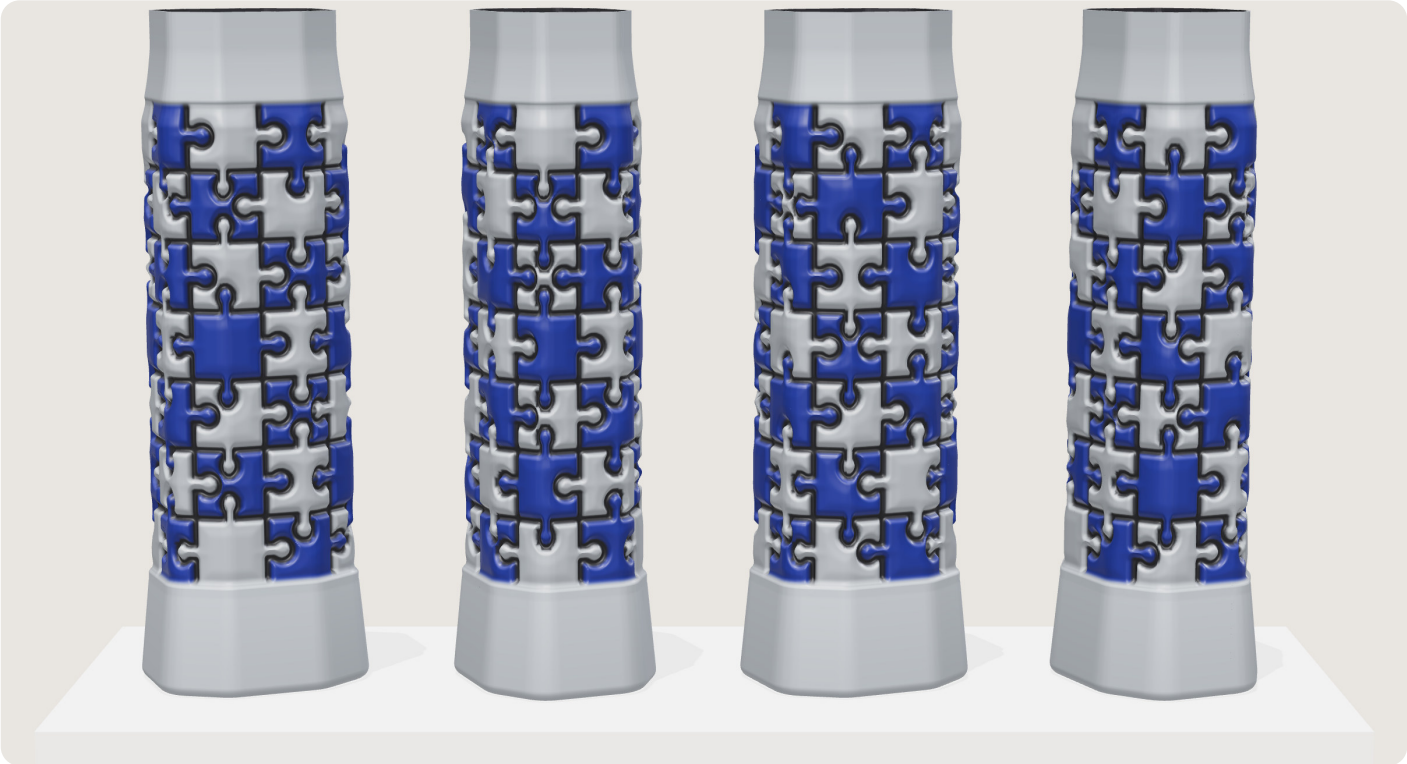
Relief floor above reference line (design / min)	**1.20 / 1.20 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**2.81 mm**

OUTER DIMENSIONS BY HEIGHT

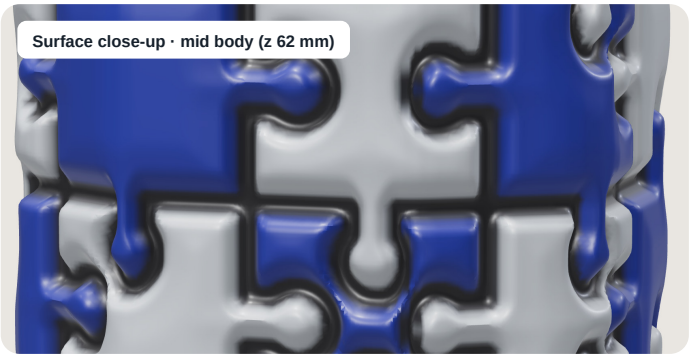
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	684	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	283	3.52	3.52
30.0	37.57	31.60	115.1	275	1.22	2.81
45.0	37.59	31.65	117.1	323	1.23	2.81
60.0	37.42	31.40	116.7	317	1.23	2.70
75.0	37.23	31.19	114.9	282	1.22	2.58
90.0	37.24	31.27	115.3	302	1.23	2.61
105.0	37.56	31.50	117.2	325	1.23	2.75
116.0	36.17	30.15	113.4	482	2.04	2.05
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.80	0.80

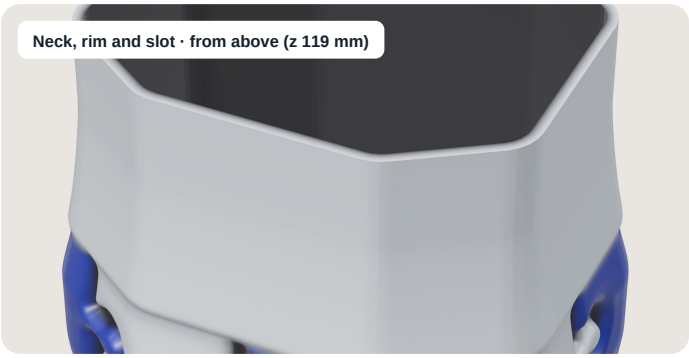
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

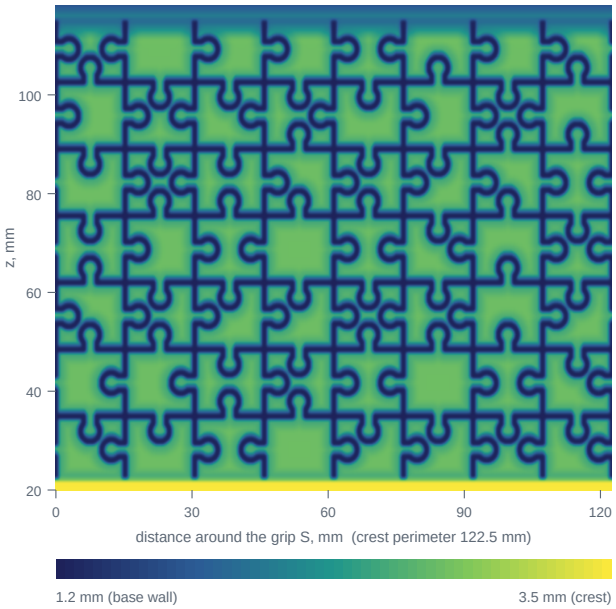


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Pieces	**8 × 7 = 56**
Piece size	**15.31 × 13.50 mm**
Knob neck / head	**2.9 mm / dia 4.2 mm**
Knob reach	**5.1 mm**
Groove width	**0.84 mm**
Groove floor	**1.20 mm (base wall)**
Piece top height	**up to 2.81 mm above base**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 1**
Wall around the cavity (min)	**1.10 mm**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Widest flat bridge	**3.7 mm**

05 NEURO-TREAD

Bio-inspired raised neural network over a dimpled adaptive base.

57.6 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE



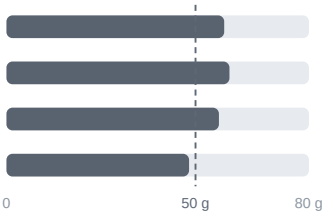
3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**39.1 × 33.0 mm**
Peak texture height	**3.50 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 3.2 counterbore**
Material volume / surface	**46.44 cm³ / 470 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**57.6 g**
PETG	1.27 g/cm³	**59.0 g**
TPU	1.21 g/cm³	**56.2 g**
ABS	1.04 g/cm³	**48.3 g**



COLOURWAY

#18181A #CE1820

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**25.2 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 19.3 m (100 % flow)**

DESIGN INTENT

A network of 48 neurons joined by curved axon ridges rises to the full 3.5 mm crest, standing proud of a field of small bumps. The ridges guide the fingers while the bump field provides fine grip.

Ridges and bumps are gentle slopes; the network is printed in the same material as the base.

KEY FEATURES

- 48 somas (8 × 6) joined by curved axons about 1.5 mm wide
- Bump field on a 2.66 mm hex pitch
- Network and somas reach the full 3.5 mm crest

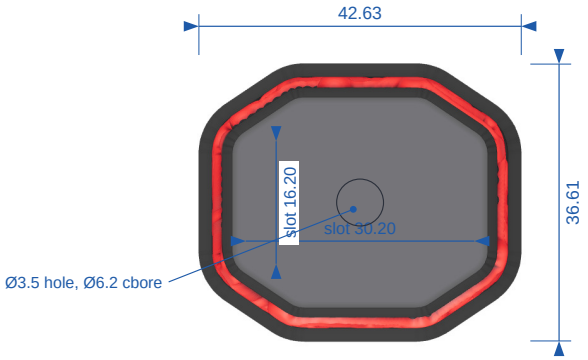
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 3.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 7**
Wall around the cavity (min)	**1.10 mm**
Material under 0.4 mm wide	**0.000 % of volume**
Widest flat bridge	**3.4 mm**

FILES

stl/05_neuro_tread.stl	24.2 MB · 483,320 triangles
3mf/05_neuro_tread.3mf	5.1 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

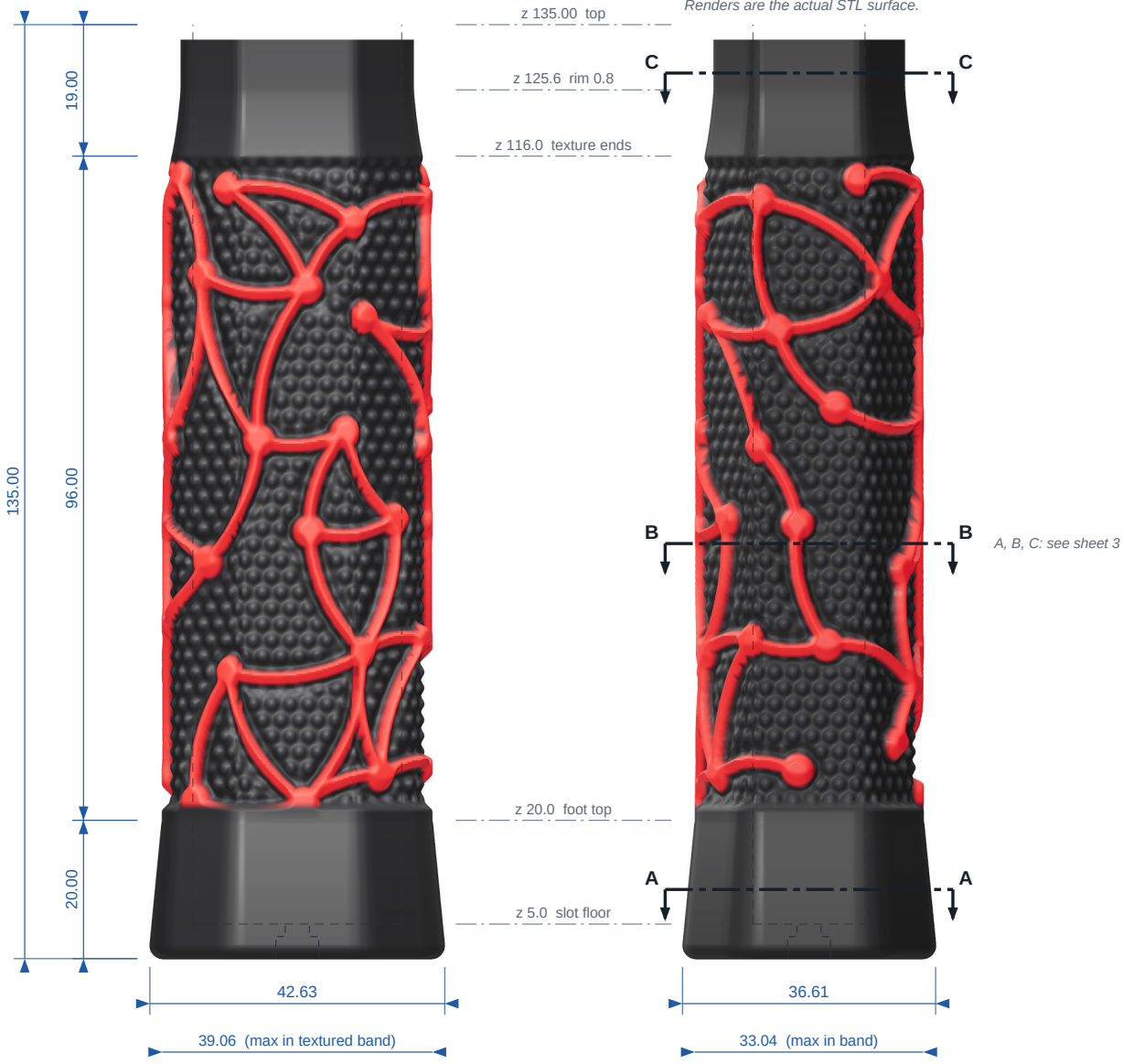


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**39.06 × 33.04**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.50**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 3.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1

0 10 20 30 40 50 mm

PART

NEURO-TREAD

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

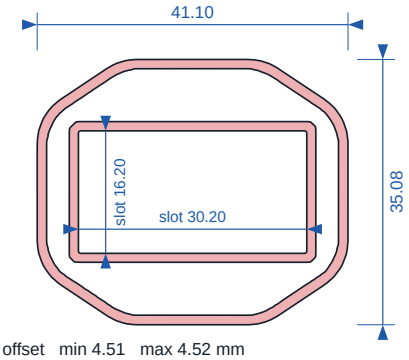
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 110 is solid.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **57.6 g**	VOLUME **46.44 cm³**	SURFACE AREA **470 cm²**
REVISION **5**	DATE **2026-10-19**	SHEET **2 of 4**

Sections & profiles

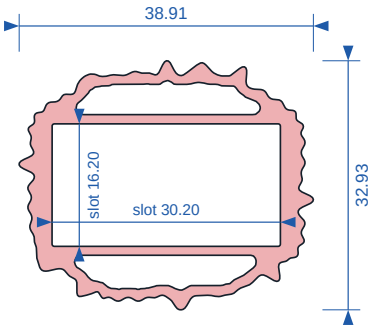
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



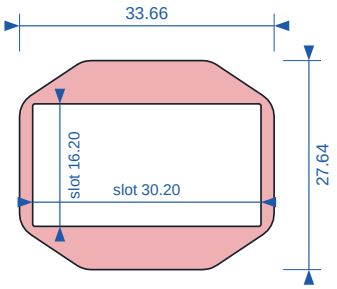
Skin offset min 4.51 max 4.52 mm
Area 278 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



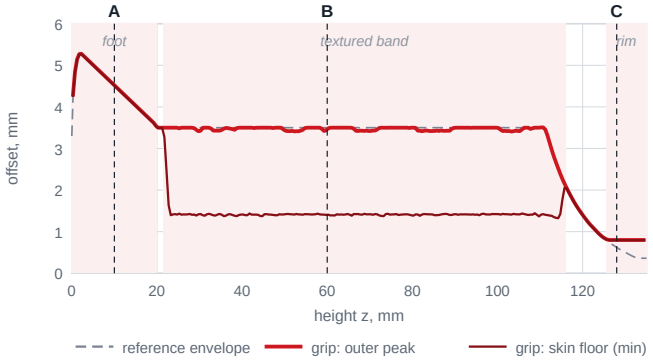
Skin offset min 1.41 max 3.46 mm
Area 303 mm² · outer perimeter 116.9 mm

SECTION C–C z = 128 mm · rim

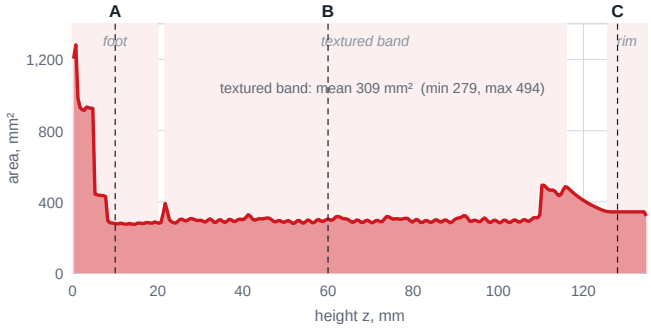


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 3.2**
Sealed cavity / min wall	**25.2 cm³ / 1.10 mm**
Outer W × D in band (max)	**39.06 × 33.04 mm**
Material volume / surface	**46.44 cm³ / 469.6 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	135.00	**9.4**

Relief floor above reference line (design / min)	**1.20 / 1.37 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.50 mm**

OUTER DIMENSIONS BY HEIGHT

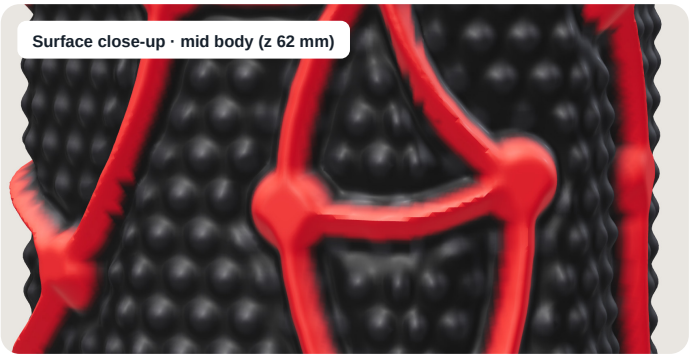
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	684	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	283	3.52	3.52
30.0	37.77	31.75	116.8	297	1.42	3.42
45.0	38.99	32.88	117.1	307	1.40	3.50
60.0	38.91	32.93	116.9	302	1.41	3.46
75.0	38.94	32.96	119.3	306	1.41	3.50
90.0	38.99	32.86	117.2	304	1.42	3.50
105.0	38.74	32.79	116.7	288	1.42	3.42
116.0	36.24	30.19	113.5	483	2.05	2.10
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.80	0.80

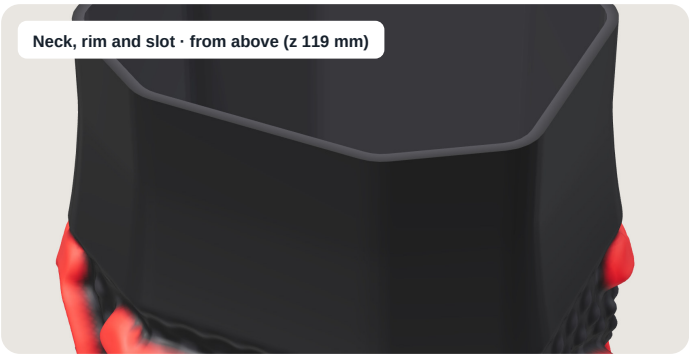
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

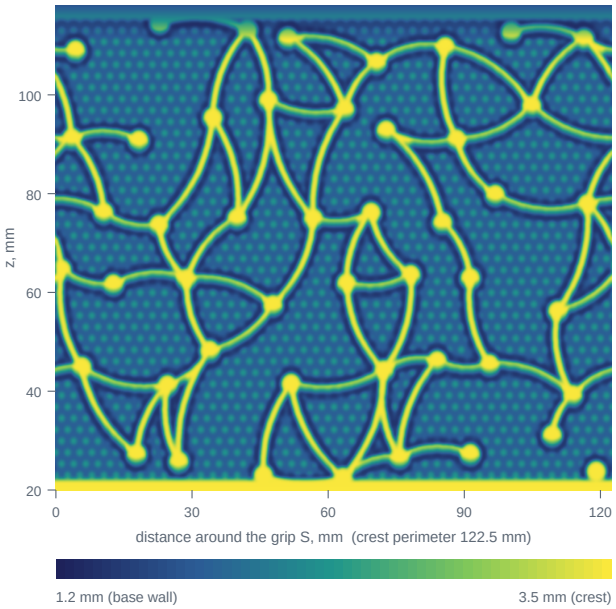


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Neurons (somas)	**48 (8 × 6)**
Axon ridge width	**about 1.5 mm**
Soma diameter	**about 4.2 mm**
Network height	**3.50 mm above base**
Bump pitch	**2.66 mm (hex)**
Bump field height	**1.4 – 2.4 mm above base**
Ridge / bump colours	**red / black**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 7**
Wall around the cavity (min)	**1.10 mm**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Widest flat bridge	**3.4 mm**

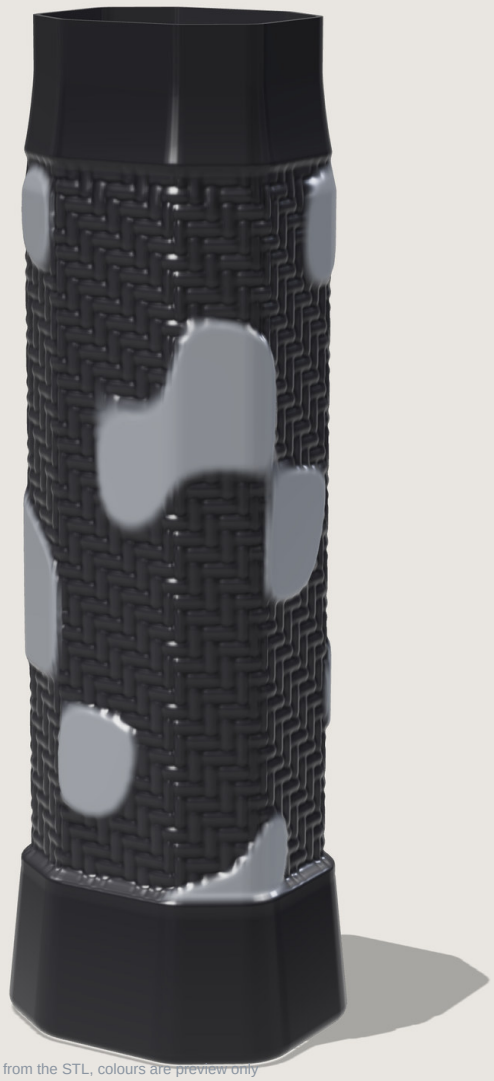
06 CARBON MATRIX

Continuous 2/2 twill weave with selective reinforcement patches.

56.4 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE

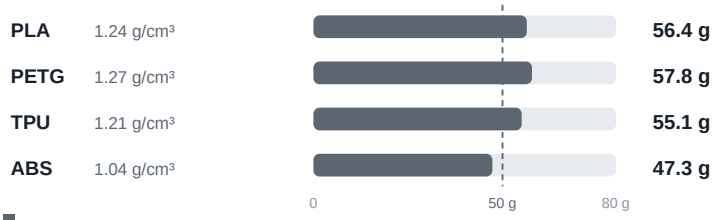


3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**37.5 × 31.4 mm**
Peak texture height	**2.73 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 3.2 counterbore**
Material volume / surface	**45.50 cm³ / 449 cm²**

WEIGHT · 100 % INFILL



COLOURWAY



#1C1C20 #80848A

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**27.4 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 18.9 m (100 % flow)**

DESIGN INTENT

A woven-carbon look built as real geometry: every tow rises over and dives under its neighbours in a 2/2 twill. Smooth, slightly raised patches suggest selective reinforcement where the palm loads the handle.

The weave is the finest texture in the set (0.4 mm relief on a 2 mm pitch).

KEY FEATURES

- 60 tows around, 2.04 mm pitch, 2/2 twill
- Weave relief about 0.4 mm; every tow is a rounded ridge
- Smooth reinforcement patches stand about 0.5 mm above the weave

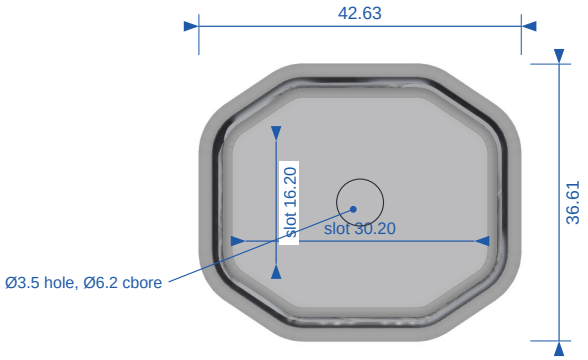
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 3.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 1**
Wall around the cavity (min)	**1.11 mm**
Material under 0.4 mm wide	**0.000 % of volume**
Widest flat bridge	**3.6 mm**

FILES

stl/06_carbon_matrix.stl	19.9 MB · 397,168 triangles
3mf/06_carbon_matrix.3mf	4.2 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

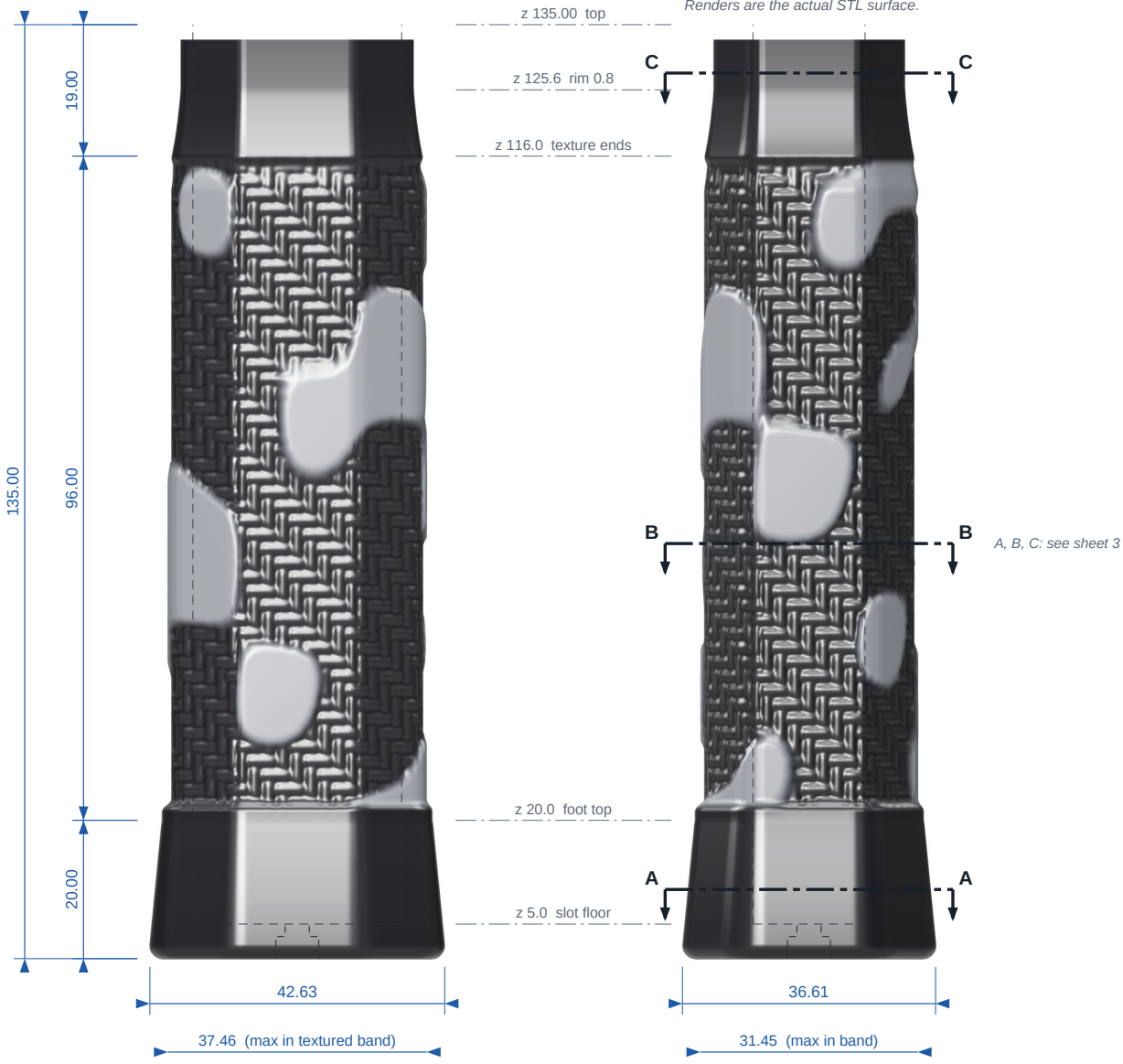


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**37.46 × 31.45**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**2.73**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 3.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

CARBON MATRIX

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

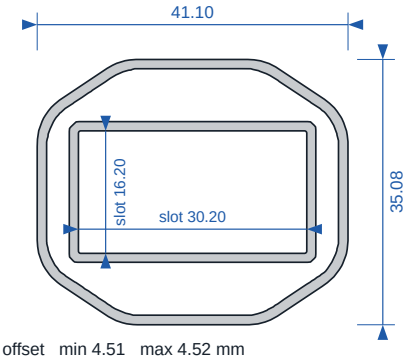
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 110 is solid.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE	UNITS	PROJECTION
1 : 1	millimetres	
MATERIAL	**INFILL**	**LAYER HEIGHT**
PLA / PETG / TPU	100 %	0.20 mm
MASS (PLA)	**VOLUME**	**SURFACE AREA**
56.4 g	45.50 cm³	449 cm²
REVISION	**DATE**	**SHEET**
5	2026-10-19	2 of 4

Sections & profiles

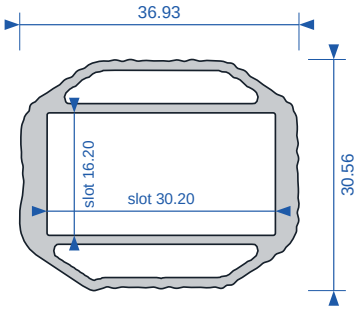
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



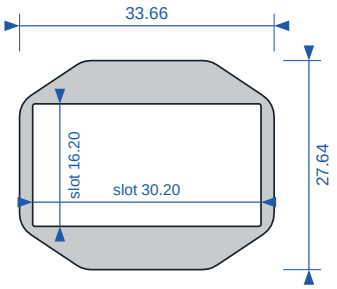
Skin offset min 4.51 max 4.52 mm
Area 278 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



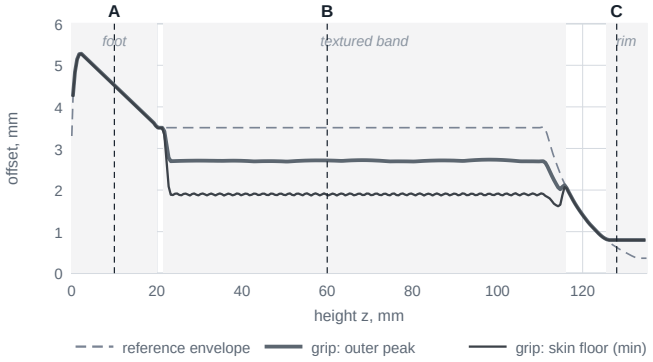
Skin offset min 1.88 max 2.71 mm
Area 287 mm² · outer perimeter 114.4 mm

SECTION C–C z = 128 mm · rim

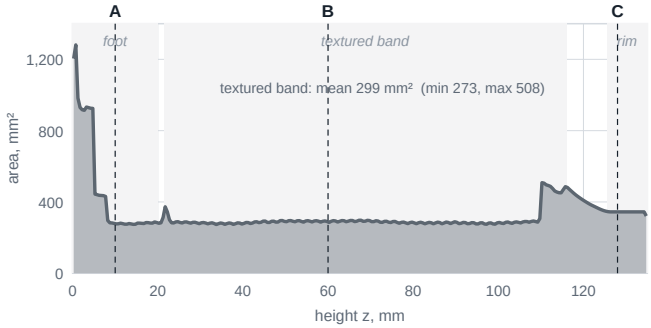


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 3.2**
Sealed cavity / min wall	**27.4 cm³ / 1.11 mm**
Outer W × D in band (max)	**37.46 × 31.45 mm**
Material volume / surface	**45.50 cm³ / 449.4 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	135.00	**9.4**

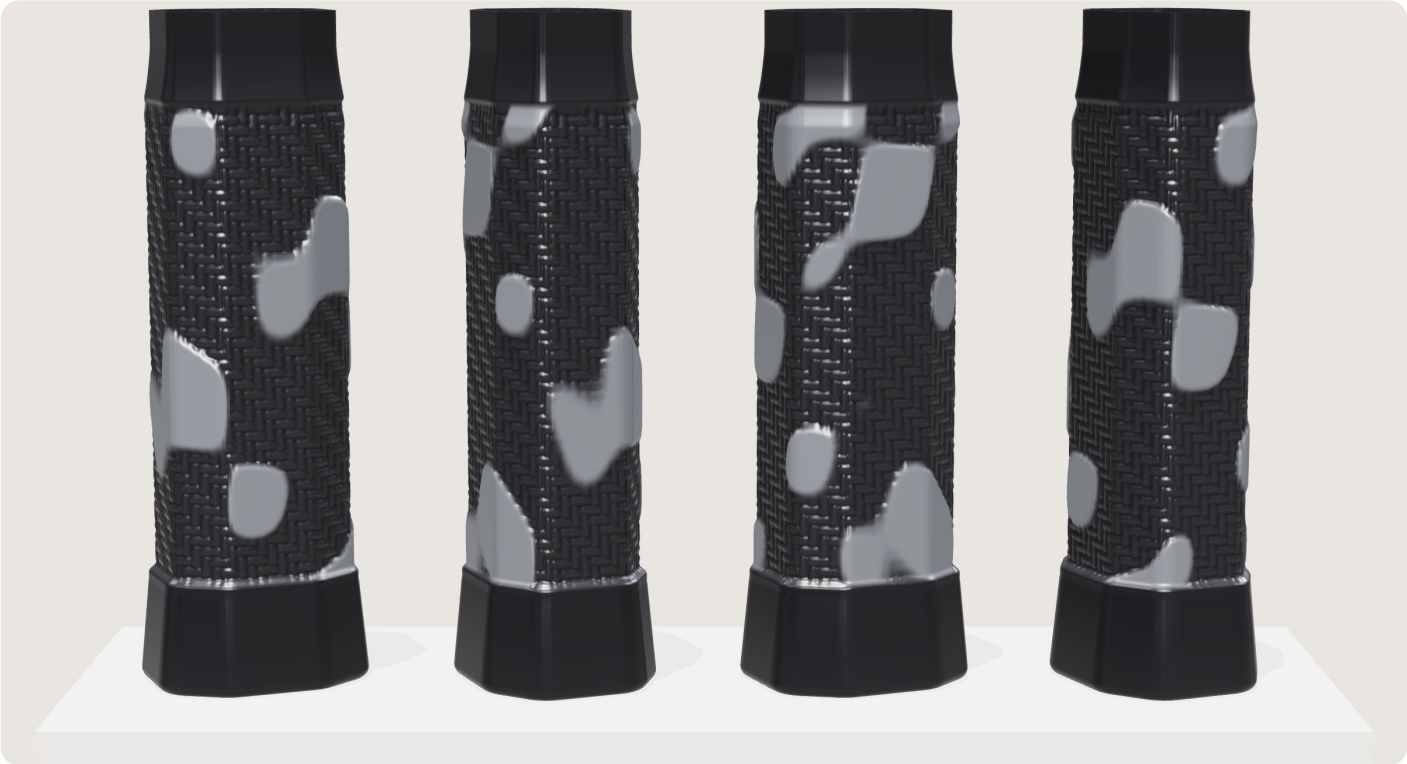
Relief floor above reference line (design / min)	**1.20 / 1.87 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**2.73 mm**

OUTER DIMENSIONS BY HEIGHT

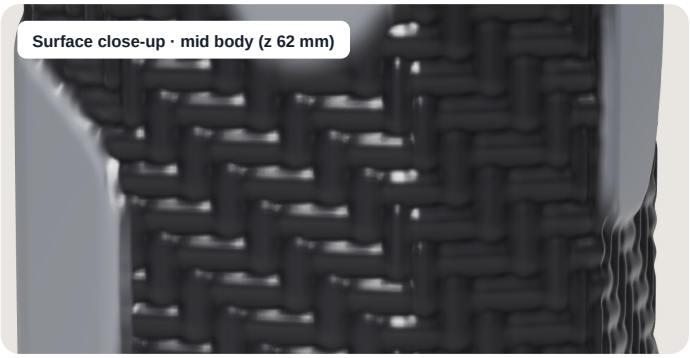
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	684	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	283	3.52	3.52
30.0	36.88	30.88	114.8	281	1.89	2.71
45.0	37.26	31.01	114.5	292	1.91	2.69
60.0	36.96	30.56	114.5	288	1.88	2.71
75.0	36.88	30.87	114.5	287	1.89	2.69
90.0	36.78	31.08	114.9	287	1.91	2.71
105.0	36.89	30.87	114.7	284	1.88	2.69
116.0	36.21	30.20	113.5	483	2.06	2.08
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.80	0.80

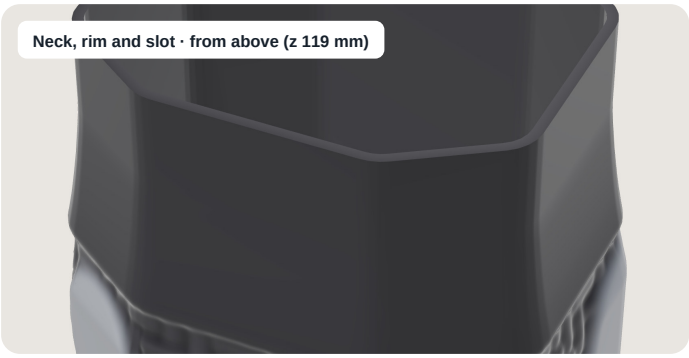
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

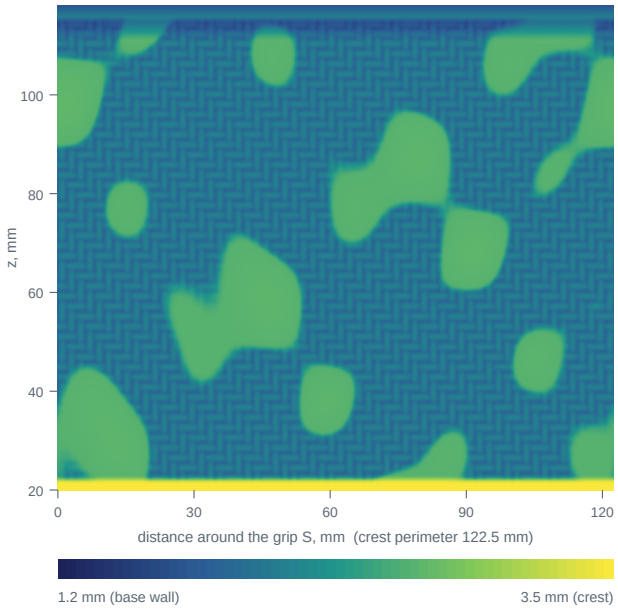


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Tows around	**60 (2/2 twill)**
Tow pitch	**2.04 mm**
Weave heights	**1.7 – 2.2 mm above base**
Patch heights	**2.3 – 2.7 mm above base**
Patch coverage	**about 23% of the band**
Weave / patch colours	**carbon / grey**
Textured band	**94.5 mm tall**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 1**
Wall around the cavity (min)	**1.11 mm**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Widest flat bridge	**3.6 mm**

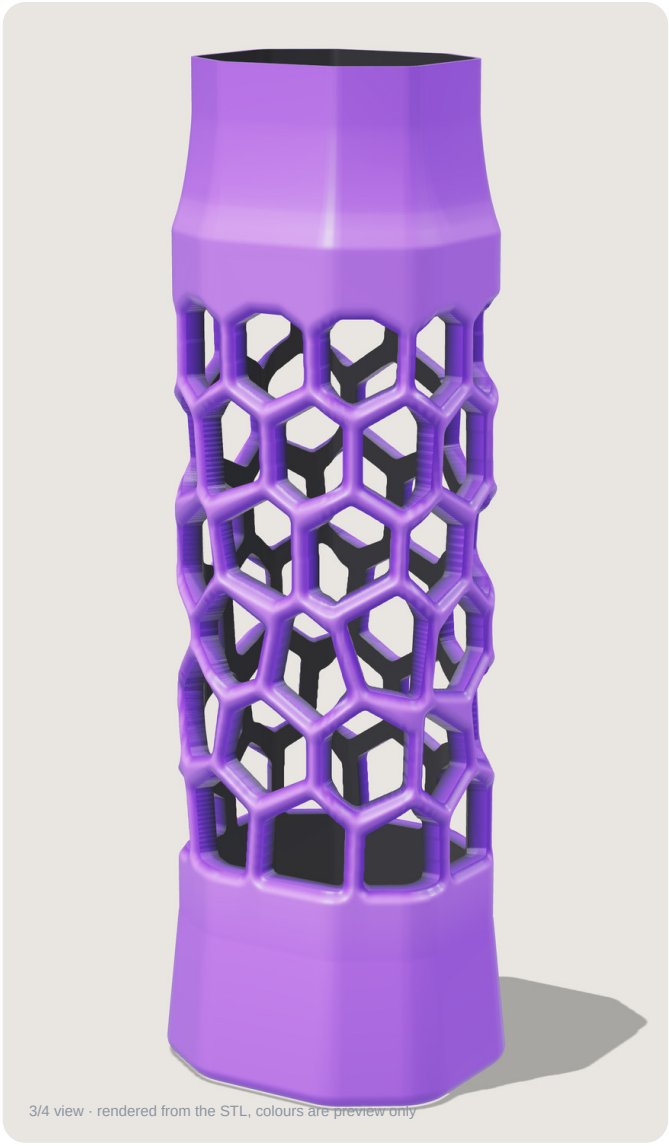
07 VORONOI-CORE

Voronoi rib lattice with recessed cells between rounded ribs.

59.5 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE



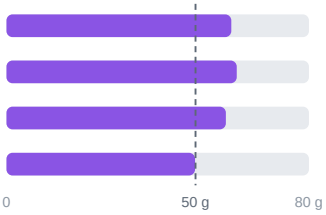
3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**39.1 × 33.0 mm**
Peak texture height	**3.50 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 3.2 counterbore**
Material volume / surface	**47.95 cm³ / 475 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**59.5 g**
PETG	1.27 g/cm³	**60.9 g**
TPU	1.21 g/cm³	**58.0 g**
ABS	1.04 g/cm³	**49.9 g**



COLOURWAY

#8A54E4

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**31.6 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 19.9 m (100 % flow)**

DESIGN INTENT

Irregular Voronoi cells become 2.6 mm ribs with rounded crests, and the cells between the ribs are recessed about 1 mm. The lattice is relaxed for even cells but keeps a natural irregularity. Like the other nine it is filled, with a sealed hollow core in the walls.

Rib flanks rise at no more than 45 degrees, so no supports are needed.

KEY FEATURES

- 54 recessed cells in the band z 27.5 – 103.5 mm
- 2.6 mm ribs standing at the full 3.5 mm crest
- Cell floors 2.46 mm above base, about 1 mm below the rib crests

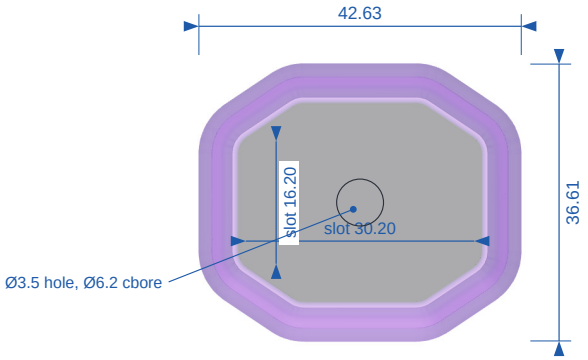
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 3.2**
Outer envelope vs reference	**+0.003 mm at most**
Outer body / sealed cavity	**1 / 6**
Wall around the cavity (min)	**1.14 mm**
Material under 0.4 mm wide	**0.001 % of volume**
Widest flat bridge	**3.7 mm**

FILES

stl/07_voronoi_core.stl	11.3 MB · 225,794 triangles
3mf/07_voronoi_core.3mf	2.4 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

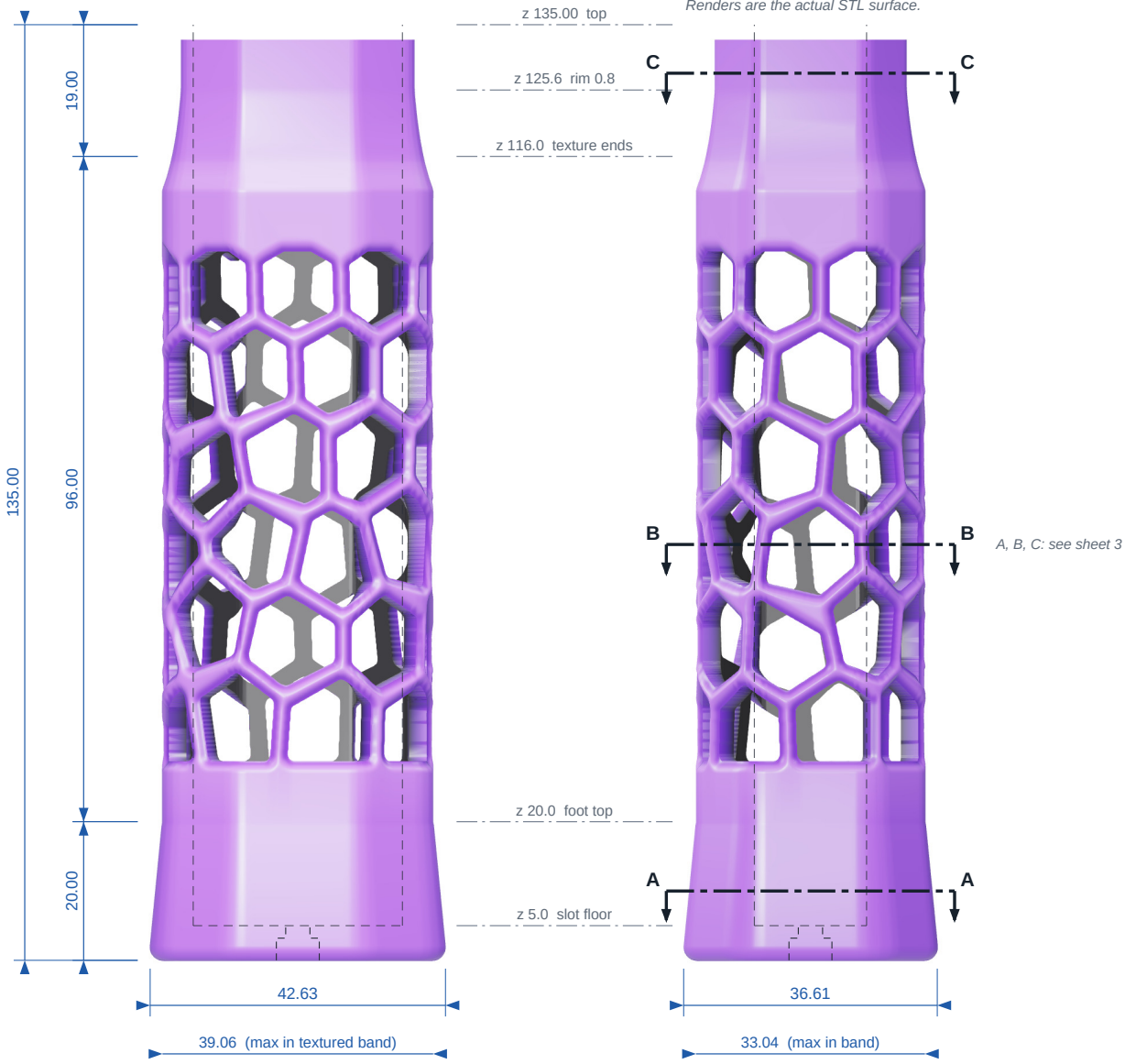


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**39.06 × 33.04**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.50**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 3.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

VORONOI-CORE

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

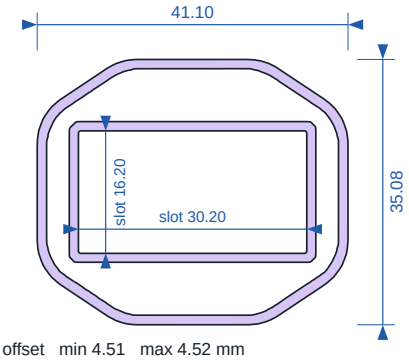
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 110 is solid.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **59.5 g**	VOLUME **47.95 cm³**	SURFACE AREA **475 cm²**
REVISION **5**	DATE **2026-10-19**	SHEET **2 of 4**

Sections & profiles

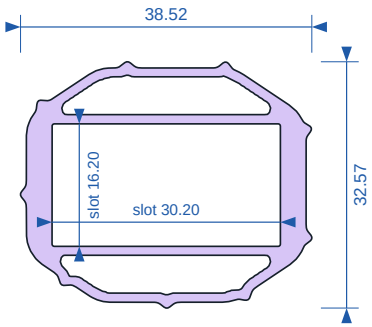
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



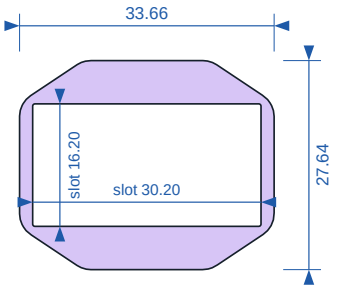
Skin offset min 4.51 max 4.52 mm
Area 278 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



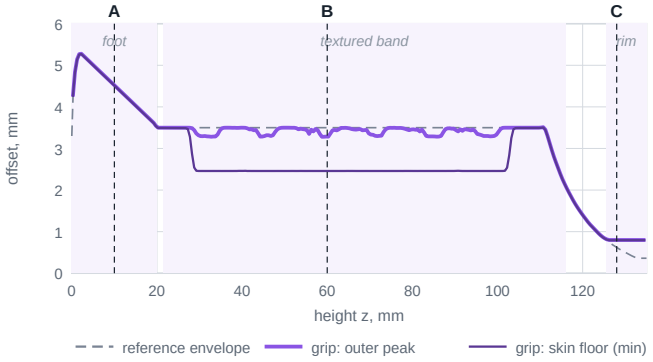
Skin offset min 2.46 max 3.29 mm
Area 303 mm² · outer perimeter 118.3 mm

SECTION C–C z = 128 mm · rim

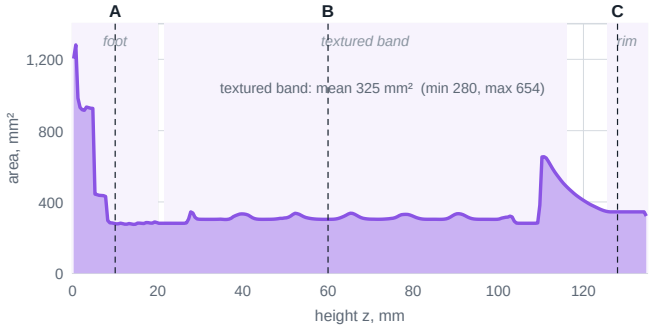


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 3.2**
Sealed cavity / min wall	**31.6 cm³ / 1.14 mm**
Outer W × D in band (max)	**39.06 × 33.04 mm**
Material volume / surface	**47.95 cm³ / 475.3 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	135.00	**9.4**

Relief floor above reference line (design / min)	**1.20 / 2.46 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.50 mm**

OUTER DIMENSIONS BY HEIGHT

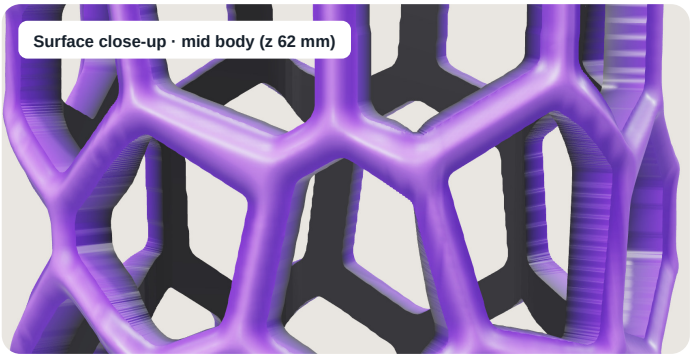
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	684	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	282	3.52	3.52
30.0	38.63	32.65	118.1	303	2.46	3.31
45.0	38.57	32.62	118.3	302	2.46	3.29
60.0	38.52	32.57	118.3	303	2.46	3.29
75.0	38.81	32.61	118.5	308	2.46	3.50
90.0	38.91	32.86	120.5	331	2.46	3.46
105.0	39.06	33.04	122.5	281	3.49	3.50
116.0	36.27	30.24	113.7	487	2.10	2.10
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.80	0.80

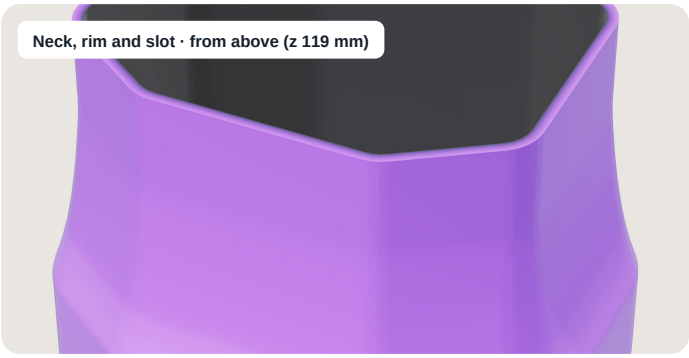
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

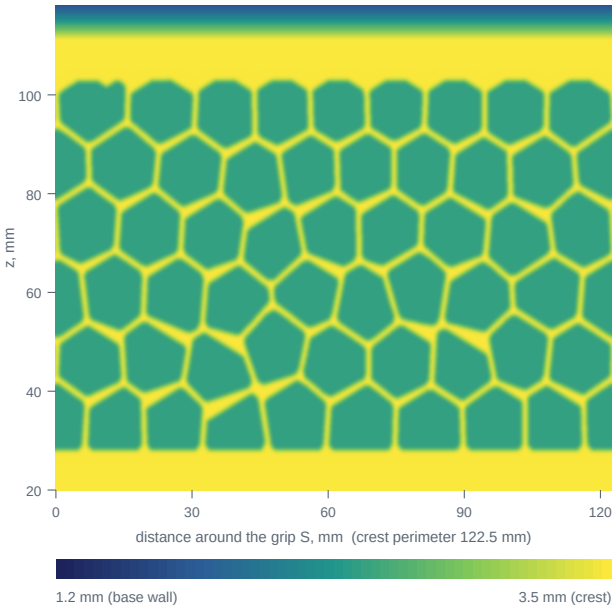


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Cells	**54, recessed about 1 mm**
Rib width	**2.6 mm**
Rib height	**3.50 mm above base**
Textured band	**z 27.5 – 103.5 mm (76 mm)**
Cell corner radius	**1.0 mm**
Cell floor	**2.46 mm above base**
Recessed area	**about 64% of the band**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 6**
Wall around the cavity (min)	**1.14 mm**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.003 mm**
Material under 0.4 mm wide	**0.001 %**
Widest flat bridge	**3.7 mm**

08 TOPO-FLOW

Continuous undulating contours mimicking wood grain and terrain.

57.8 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE

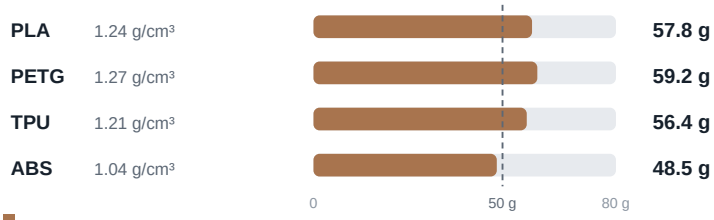


3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**38.3 × 32.1 mm**
Peak texture height	**3.13 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 3.2 counterbore**
Material volume / surface	**46.62 cm³ / 461 cm²**

WEIGHT · 100 % INFILL



COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**26.0 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 19.4 m (100 % flow)**

DESIGN INTENT

The surface is a contour map of terrain with four wood-like knots. Ridges follow the contours so the lines flow around the handle like grain in a cut log, giving directional grip that is softer on the hand than sharp knurling.

Ridge flanks are rounded and below 45 degrees; the grain is geometry, not colour alone.

KEY FEATURES

- Four knots, contour ridges spaced about 2.5 – 7 mm
- Rounded ridges: crests about 3.1 mm, valleys 1.66 mm above the base
- Three-tone wood colourway (light / mid / dark)

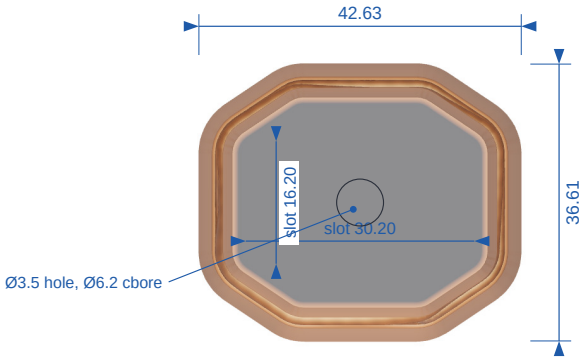
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 3.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 1**
Wall around the cavity (min)	**1.09 mm**
Material under 0.4 mm wide	**0.000 % of volume**
Widest flat bridge	**3.4 mm**

FILES

stl/08_topo_flow.stl	21.7 MB · 433,616 triangles
3mf/08_topo_flow.3mf	4.7 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

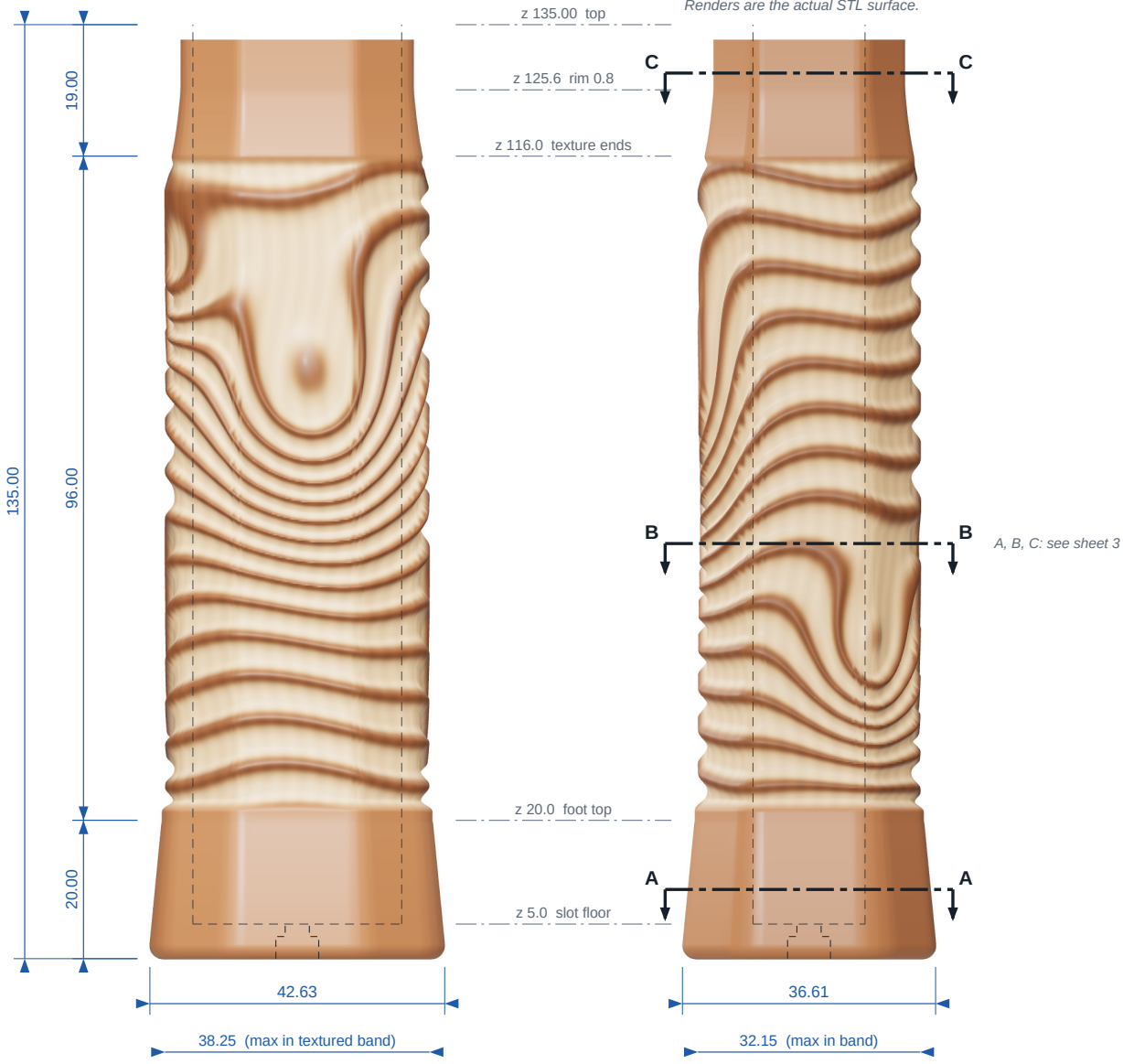


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**38.25 × 32.15**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.13**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 3.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

TOPO-FLOW

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

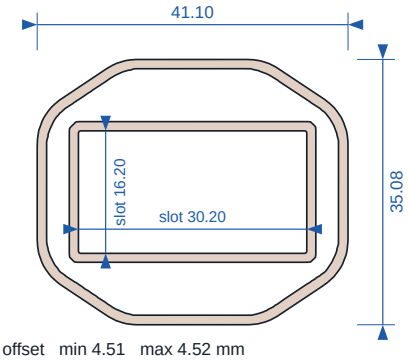
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 110 is solid.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE	UNITS	PROJECTION
1 : 1	millimetres	
MATERIAL	**INFILL**	**LAYER HEIGHT**
PLA / PETG / TPU	100 %	0.20 mm
MASS (PLA)	**VOLUME**	**SURFACE AREA**
57.8 g	46.62 cm³	461 cm²
REVISION	**DATE**	**SHEET**
5	2026-10-19	2 of 4

Sections & profiles

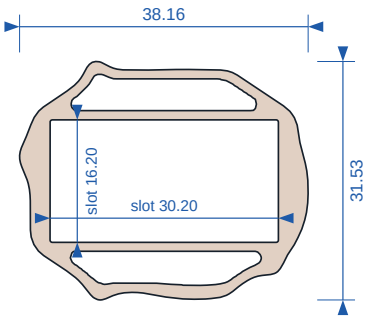
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



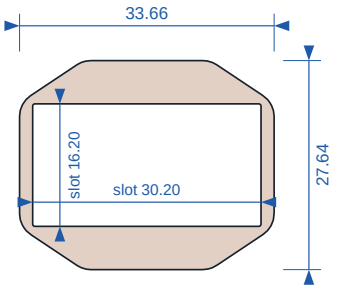
Skin offset min 4.51 max 4.52 mm
Area 278 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



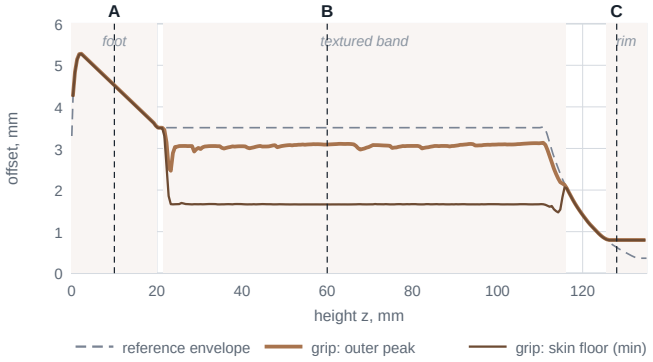
Skin offset min 1.65 max 3.09 mm
Area 291 mm² · outer perimeter 114.1 mm

SECTION C–C z = 128 mm · rim

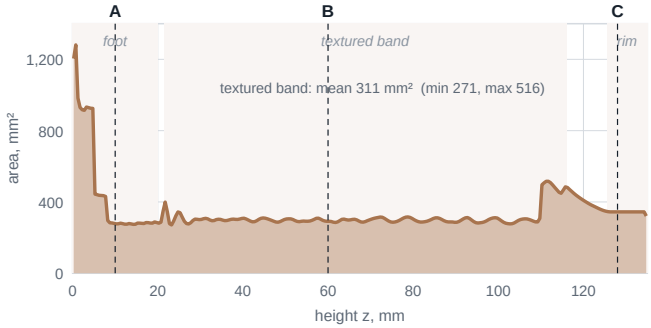


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 3.2**
Sealed cavity / min wall	**26.0 cm³ / 1.09 mm**
Outer W × D in band (max)	**38.25 × 32.15 mm**
Material volume / surface	**46.62 cm³ / 460.8 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	135.00	**9.4**

Relief floor above reference line (design / min)	**1.20 / 1.65 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.13 mm**

OUTER DIMENSIONS BY HEIGHT

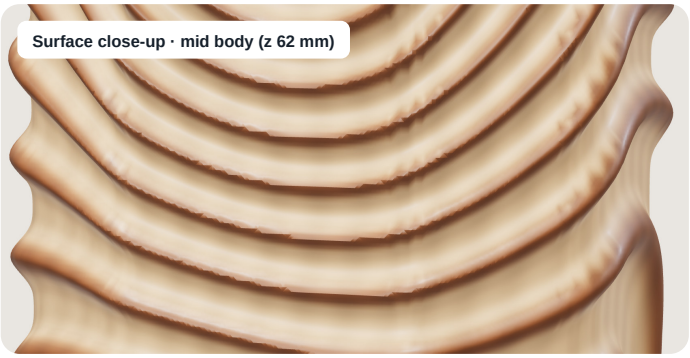
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	684	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	283	3.52	3.52
30.0	37.98	31.02	113.6	301	1.66	3.00
45.0	38.00	32.05	117.4	309	1.66	3.05
60.0	38.13	31.60	114.1	291	1.65	3.09
75.0	37.02	31.88	115.4	287	1.66	3.04
90.0	37.70	31.13	115.4	286	1.66	3.06
105.0	38.18	29.98	116.0	299	1.66	3.11
116.0	36.19	30.19	113.4	483	2.05	2.09
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.80	0.80

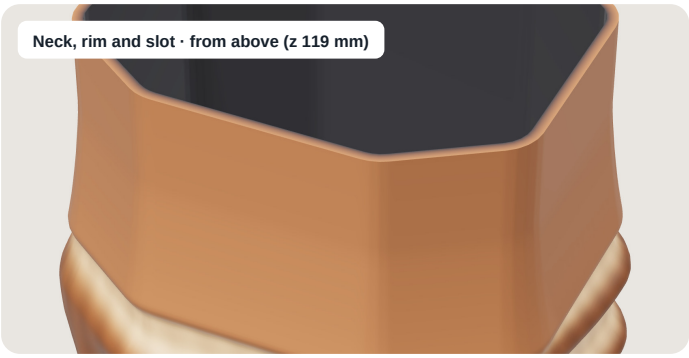
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

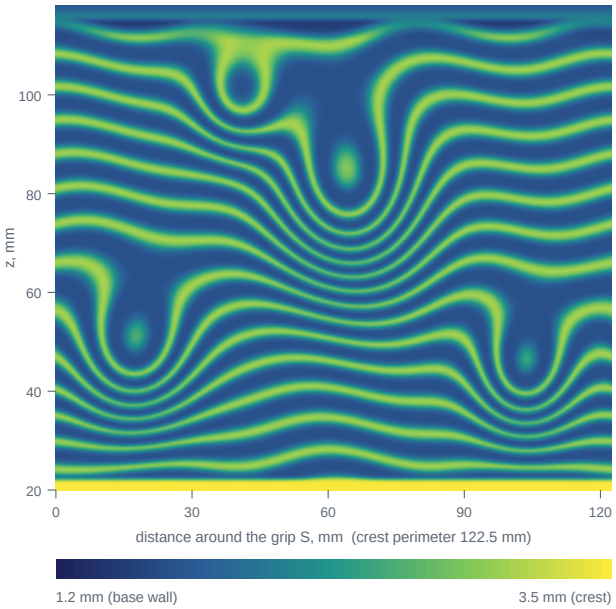


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Knots	**4**
Ridge spacing	**about 2.5 – 7 mm**
Ridge crest	**about 3.1 mm above base**
Valley floor	**1.66 mm above base**
Relief	**about 1.4 mm**
Textured band	**z 21.5 – 116.0 mm**
Colourway	**3 wood tones**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 1**
Wall around the cavity (min)	**1.09 mm**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Widest flat bridge	**3.4 mm**

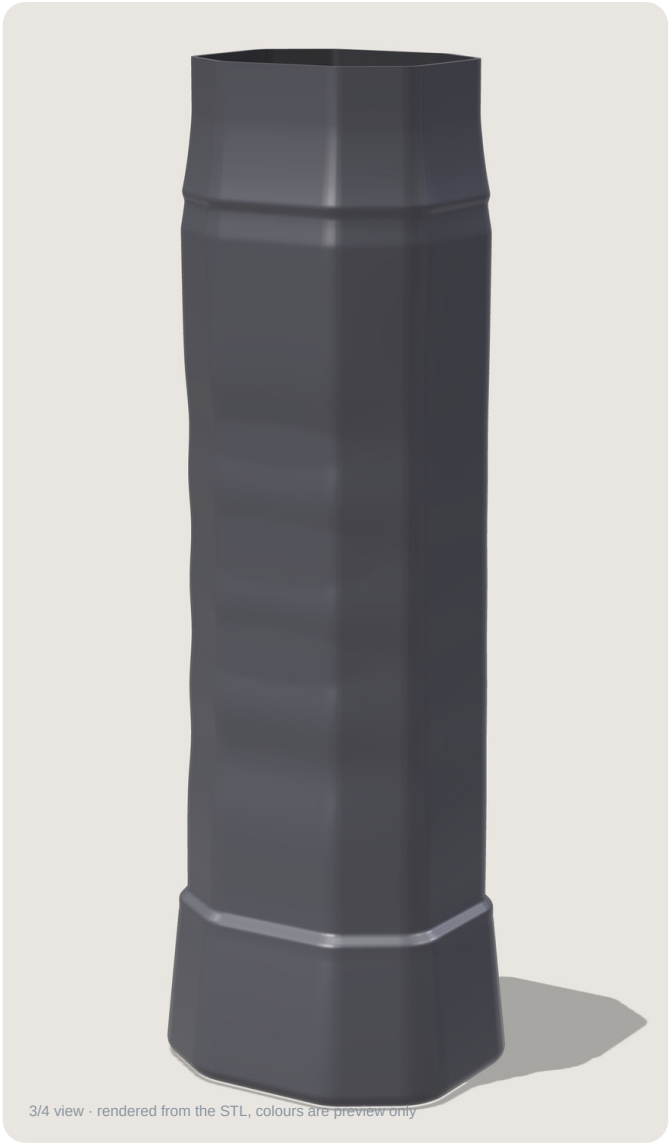
09 ERGO-CONTOUR

Palm swell, four finger flutes and a thumb rest for hand conformance.

55.5 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE

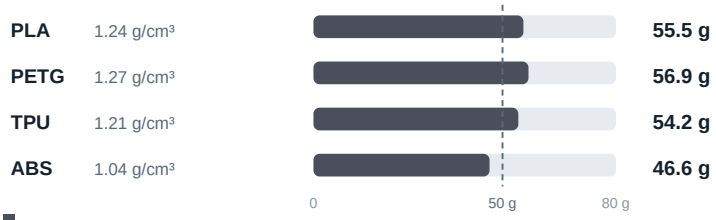


3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**36.9 × 30.9 mm**
Peak texture height	**2.77 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 3.2 counterbore**
Material volume / surface	**44.78 cm³ / 444 cm²**

WEIGHT · 100 % INFILL



COLOURWAY

#424248

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**27.2 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 18.6 m (100 % flow)**

DESIGN INTENT

A smooth grip shaped for the hand rather than textured: an hourglass swell, a raised palm pad on the back, four finger flutes and a thumb dish on the front. There are no sharp edges anywhere on the surface.

The smoothest surface in the set; it prints with the least visible layer texture.

VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 3.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 1**
Wall around the cavity (min)	**1.11 mm**
Material under 0.4 mm wide	**0.000 % of volume**
Widest flat bridge	**3.6 mm**

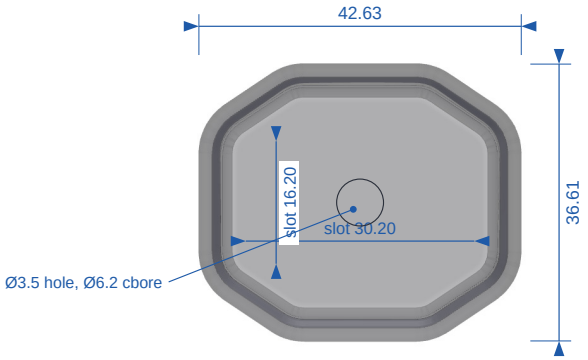
KEY FEATURES

- Four finger flutes at z = 44, 58, 72, 86 mm
- Thumb dish at z about 97 mm, palm pad at z about 62 mm
- Hourglass swell 1.73 – 2.33 mm, peaking at about 2.8 mm on the palm pad

FILES

stl/09_ergo_contour.stl	4.5 MB · 89,324 triangles
3mf/09_ergo_contour.3mf	0.9 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

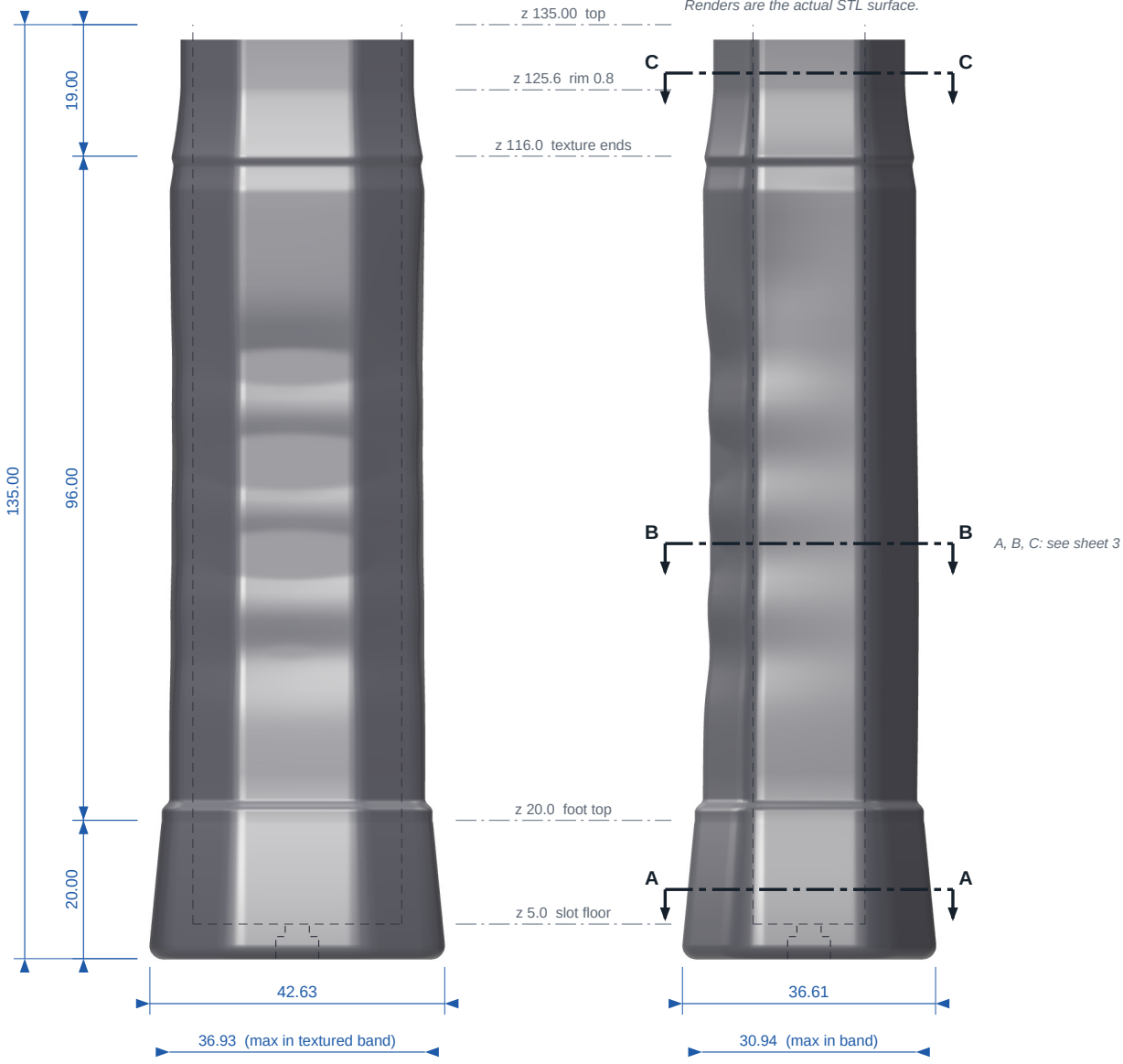


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**36.93 × 30.94**
Foot top (z 20)	**39.15 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**2.77**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 3.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

ERGO-CONTOUR

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

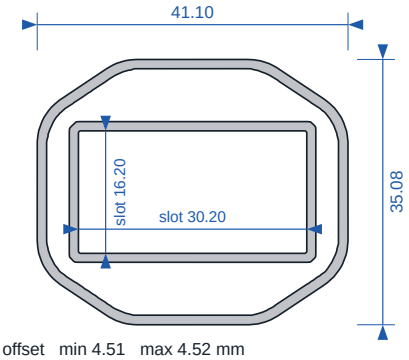
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 110 is solid.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **55.5 g**	VOLUME **44.78 cm³**	SURFACE AREA **444 cm²**
REVISION **5**	DATE **2026-10-19**	SHEET **2 of 4**

Sections & profiles

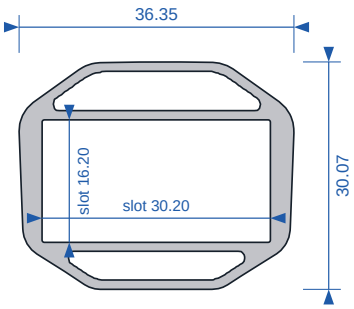
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



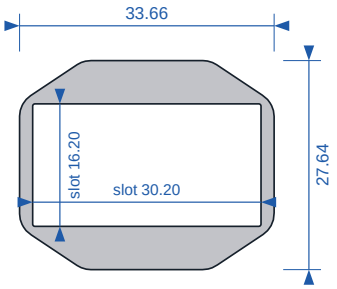
Skin offset min 4.51 max 4.52 mm
Area 278 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



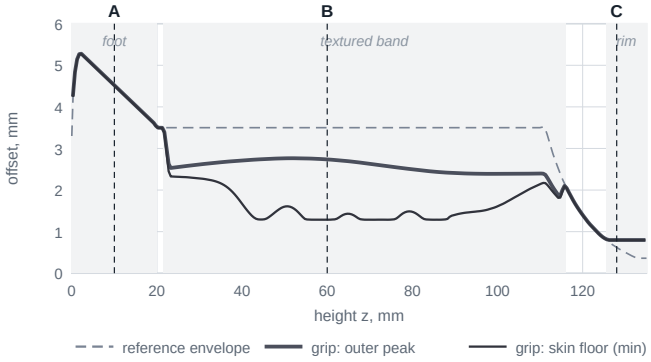
Skin offset min 1.29 max 2.74 mm
Area 269 mm² · outer perimeter 112.6 mm

SECTION C–C z = 128 mm · rim

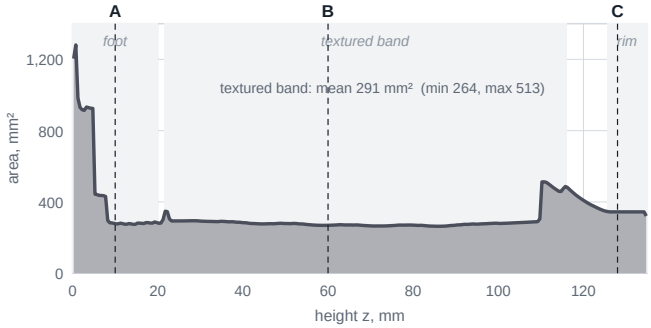


Skin offset min 0.80 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 3.2**
Sealed cavity / min wall	**27.2 cm³ / 1.11 mm**
Outer W × D in band (max)	**36.93 × 30.94 mm**
Material volume / surface	**44.78 cm³ / 444.2 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	135.00	**9.4**

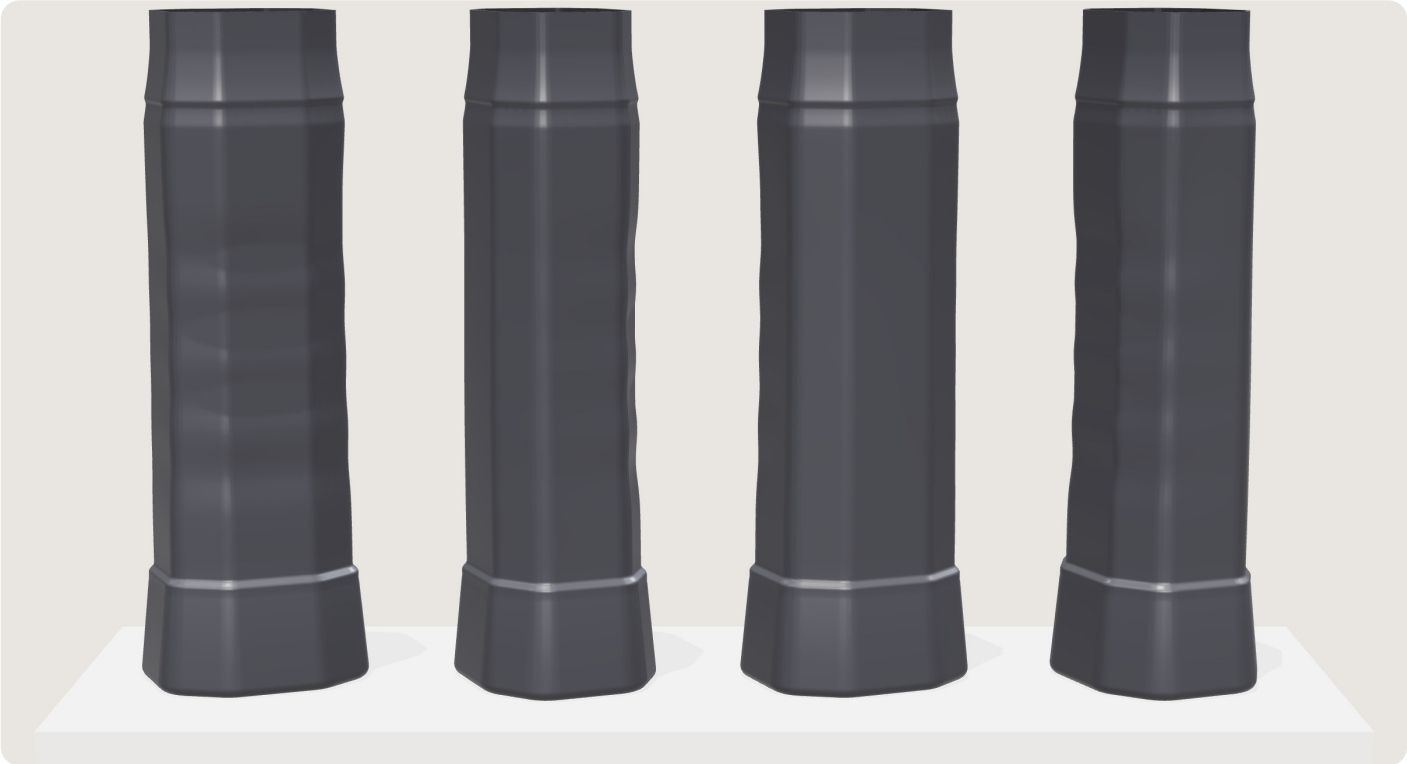
Relief floor above reference line (design / min)	**1.20 / 1.29 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**2.77 mm**

OUTER DIMENSIONS BY HEIGHT

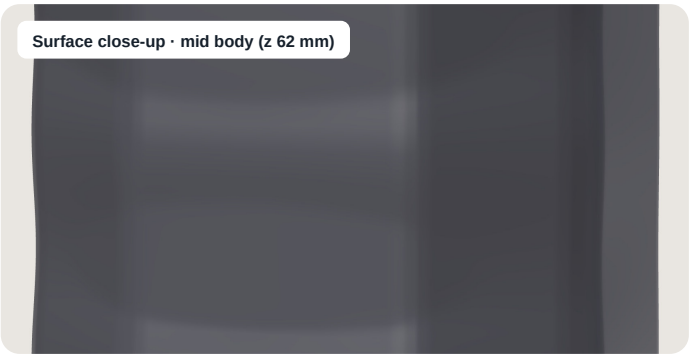
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	684	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	282	3.52	3.52
30.0	36.93	30.94	115.6	293	2.28	2.62
45.0	36.60	30.16	113.3	277	1.30	2.74
60.0	36.35	30.07	112.6	269	1.29	2.74
75.0	36.18	29.90	112.2	267	1.29	2.56
90.0	36.27	30.01	112.5	270	1.40	2.41
105.0	36.66	30.68	114.6	284	1.88	2.39
116.0	36.21	30.19	113.5	484	2.07	2.08
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.80	0.80

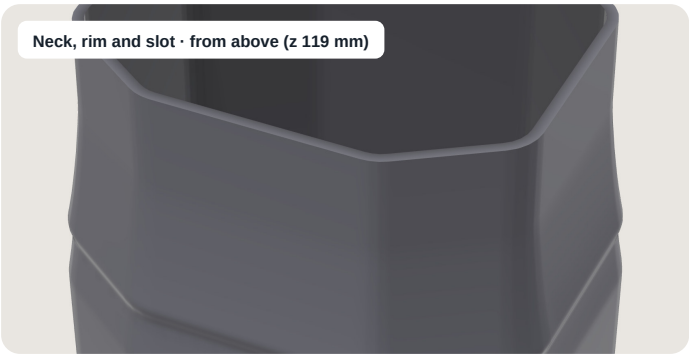
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

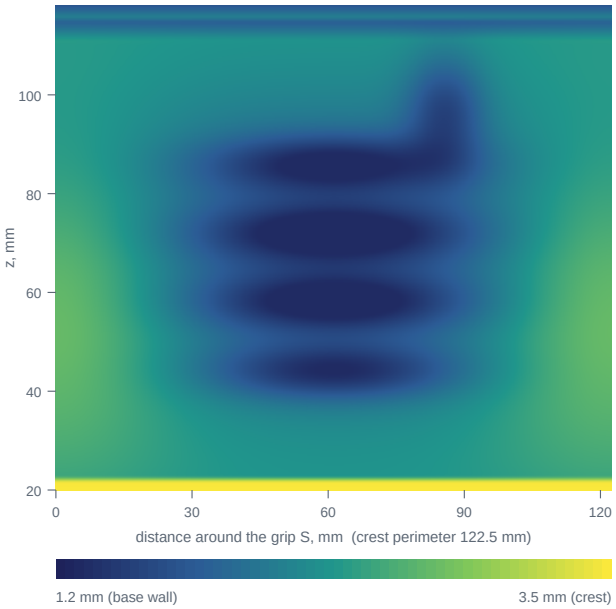


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Finger flutes	**4, at z 44 / 58 / 72 / 86**
Flute width / depth	**about 9 mm / 0.8 mm**
Thumb dish	**z 97 mm, depth 0.6 mm**
Palm pad	**back face, z 62 mm, +1.0 mm**
Hourglass swell	**1.73 – 2.33 mm above base**
Peak height	**about 2.8 mm above base**
Surface	**smooth, no sharp edges**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 1**
Wall around the cavity (min)	**1.11 mm**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Widest flat bridge	**3.6 mm**

10 HEXA-MOD

Modular hex pads on four height levels, colour-coded (red = tallest).

57.5 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE

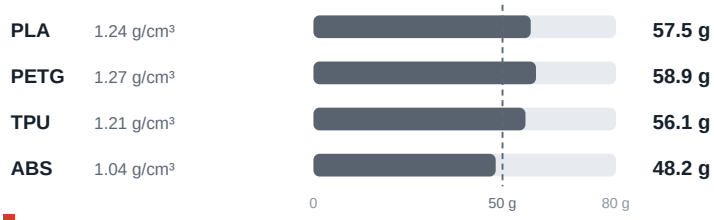


3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**38.7 × 33.0 mm**
Peak texture height	**3.50 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 3.2 counterbore**
Material volume / surface	**46.34 cm³ / 471 cm²**

WEIGHT · 100 % INFILL



COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**25.7 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 19.3 m (100 % flow)**

DESIGN INTENT

A honeycomb of individual hex pads, each separated by a 1.7 mm gap and set at one of four heights. The height map is pseudo-random with a gentle swell along the handle; the tallest pads (red) reach the full 3.5 mm crest.

Pads are chamfered on the lower edge so they never overhang beyond 45 degrees.

KEY FEATURES

- 12 columns × 11 rows of pads, 10.2 mm across flats
- Four levels: 2.35 / 2.76 / 3.13 / 3.50 mm above the base
- Red pads mark the tallest level (about 6% of the band)

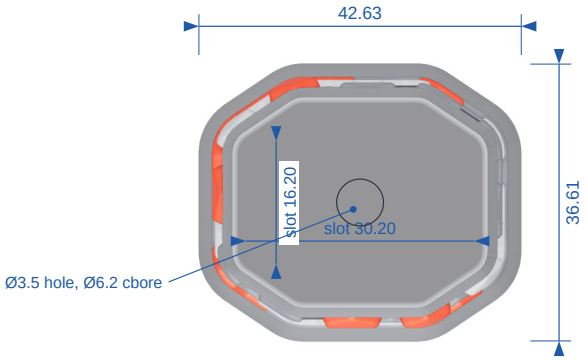
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 3.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 3**
Wall around the cavity (min)	**1.09 mm**
Material under 0.4 mm wide	**0.001 % of volume**
Widest flat bridge	**3.6 mm**

FILES

stl/10_hexa_mod.stl	17.0 MB · 340,206 triangles
3mf/10_hexa_mod.3mf	3.6 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

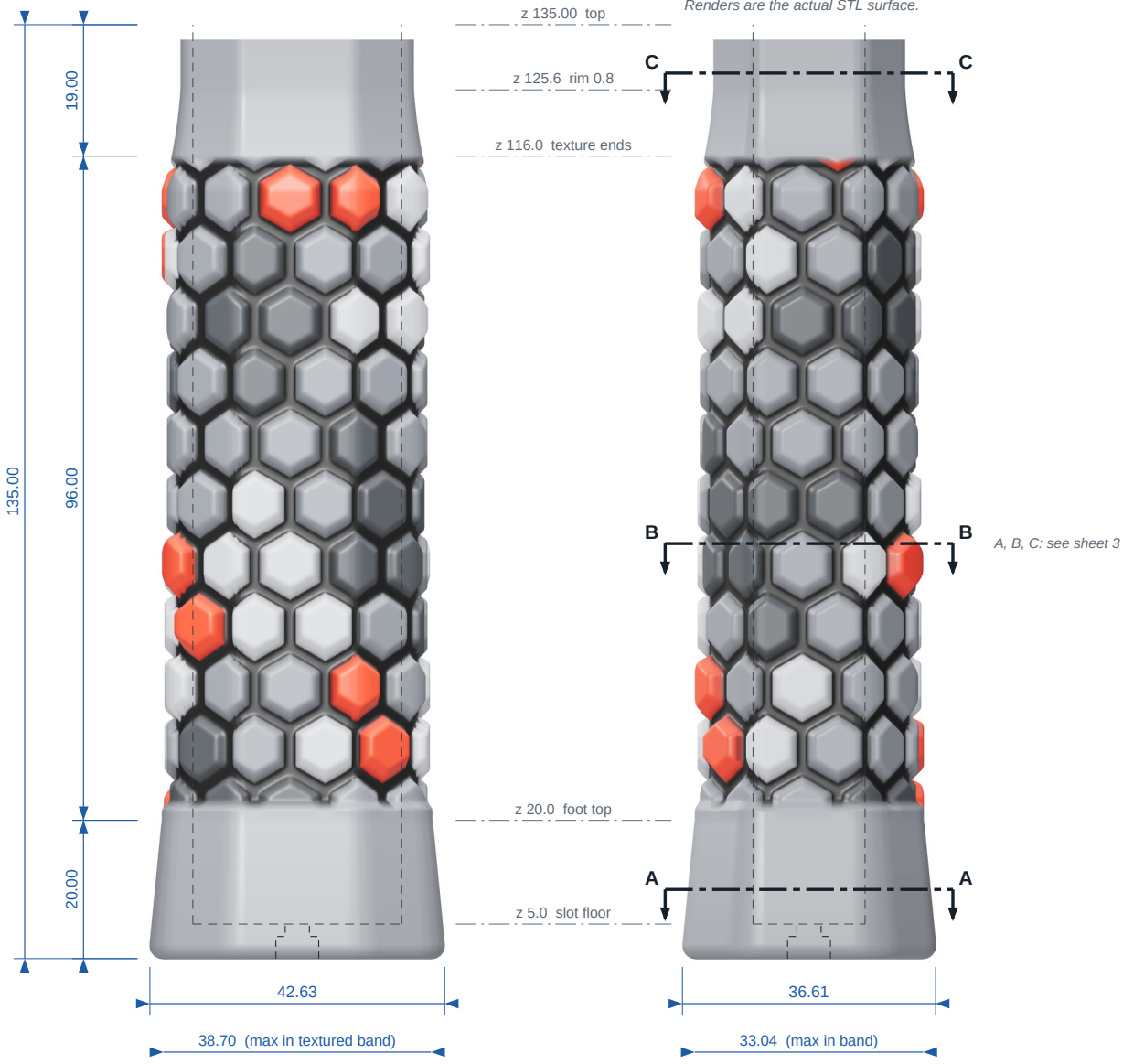


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**38.70 × 33.04**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.50**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 3.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

HEXA-MOD

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

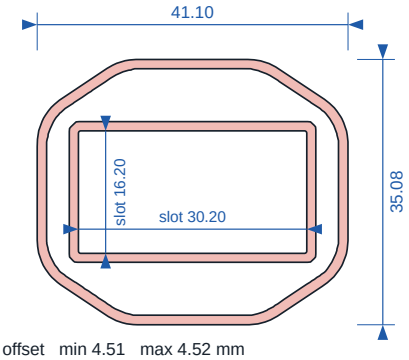
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 110 is solid.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **57.5 g**	VOLUME **46.34 cm³**	SURFACE AREA **471 cm²**
REVISION **5**	DATE **2026-10-19**	SHEET **2 of 4**

Sections & profiles

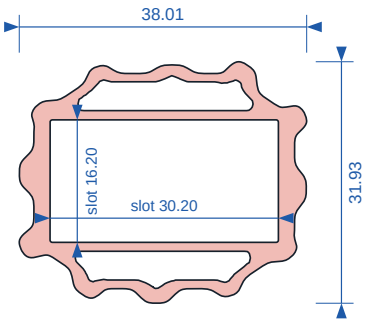
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



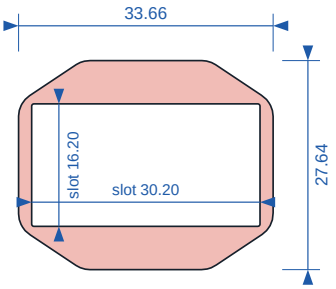
Skin offset min 4.51 max 4.52 mm
Area 278 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



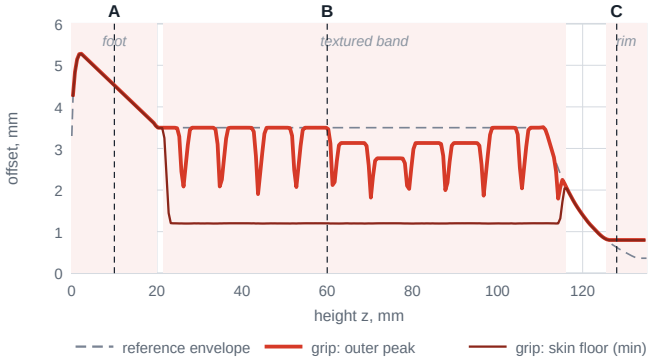
Skin offset min 1.20 max 3.48 mm
Area 298 mm² · outer perimeter 117.7 mm

SECTION C–C z = 128 mm · rim

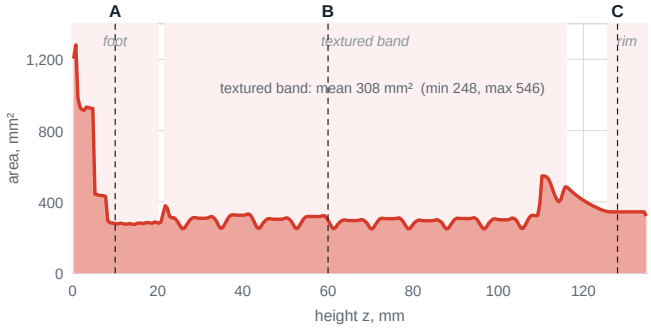


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 3.2**
Sealed cavity / min wall	**25.7 cm³ / 1.09 mm**
Outer W × D in band (max)	**38.70 × 33.04 mm**
Material volume / surface	**46.34 cm³ / 471.2 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	135.00	**9.4**

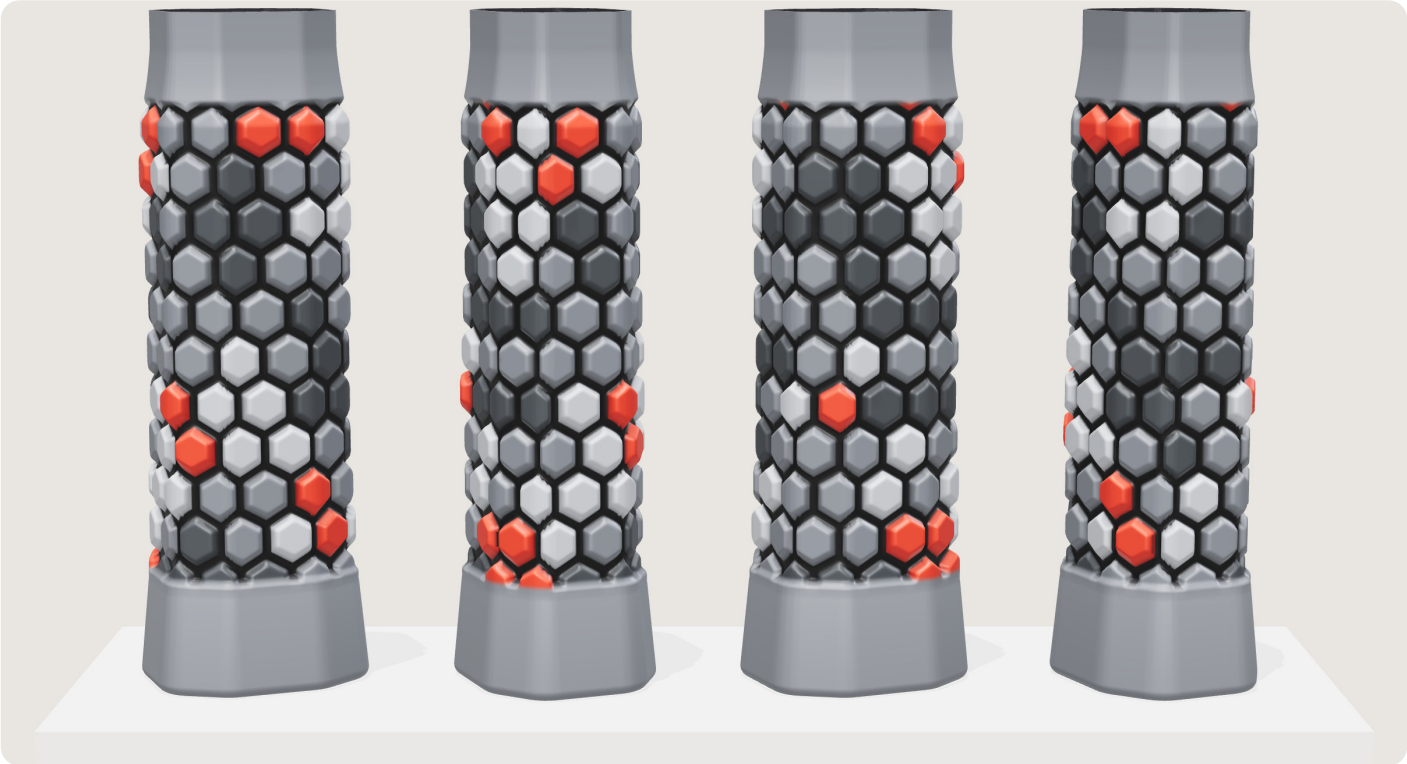
Relief floor above reference line (design / min)	**1.20 / 1.19 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.50 mm**

OUTER DIMENSIONS BY HEIGHT

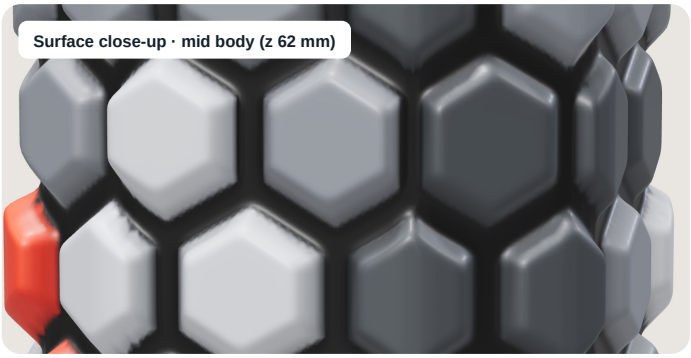
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	684	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	283	3.52	3.52
30.0	38.33	32.67	118.8	309	1.20	3.50
45.0	37.80	32.03	115.1	289	1.20	3.13
60.0	38.07	31.91	117.6	297	1.20	3.48
75.0	37.59	31.49	117.0	306	1.20	2.76
90.0	37.15	32.20	117.7	305	1.20	3.13
105.0	37.70	31.09	112.9	265	1.20	2.94
116.0	36.26	30.24	113.4	483	2.04	2.10
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.80	0.80

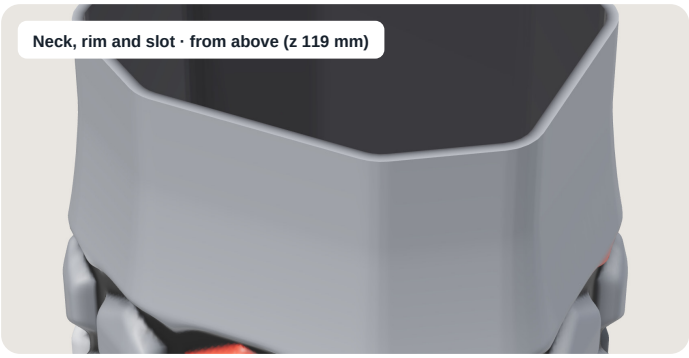
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

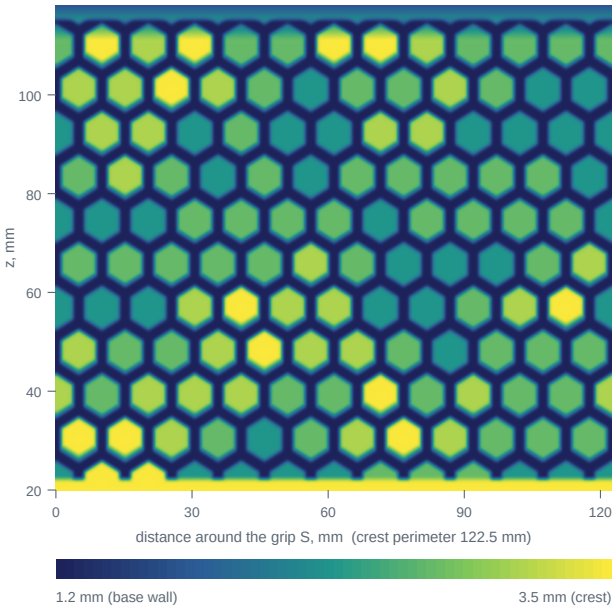


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Pad grid	**12 × 11 (pointy-top hex)**
Pad across flats	**10.21 mm**
Gap between pads	**1.7 mm**
Levels	**2.35 / 2.76 / 3.13 / 3.50 mm**
Gap floor	**1.20 mm (base wall)**
Pad edge	**chamfered, about 0.8 mm**
Level colours	**3 greys + red**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 3**
Wall around the cavity (min)	**1.09 mm**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.001 %**
Widest flat bridge	**3.6 mm**

Notes, method and file manifest

What the numbers mean, what was assumed, and what is in the package.

ABOUT THESE SHEETS

- All dimensions and volumes are measured directly from the STL files; design-intent values are only used for descriptions.
- Weight = (outer volume − sealed core) × density at 100 % infill. Slicers land within a few percent, so budget ±3 %.
- “Above base” heights are measured outward from the reference bore line (the 32.06 × 26.04 mm octagon of PickleballGrip_1.3mf), which is now inside the body.
- Assumed: 0.4 mm nozzle, 1.75 mm filament, printed upright with the butt on the bed. Not sliced or printed yet.
- Renders and colours are a preview. The STLs are single-material; the 3MFs carry a small print palette per triangle.
- Sheet numbering: PG-01 to PG-10. Drawings are third-angle, true size at 100 % print scale.

DENSITIES USED (G/CM³)

PLA	1.24
PETG	1.27
TPU	1.21
ABS	1.04

FILES

stl/	ten single-material STLs
3mf/	ten colour 3MFs (materials extension)
specs/	ten sheets + this book
renders/	catalog renders and fitment overlay
scripts/	generator, verifiers, PDF builder

REVISION HISTORY

REV	CHANGE
1	Initial ten designs, STL + colour 3MF
2	Smooth finish, solid 1.2 mm walls, ≤ 50 g each, specification sheets
3	Solid body, 30 × 16 mm paddle slot, Ø3.5 mm screw hole; Voronoi windows removed
4	Sealed hollow core (≈ 50 g), 90° countersink on the screw hole
5	135 mm long (130 mm slot, 5 mm base), cap-head counterbore, solid head above z 110, 0.5 mm slot lead-in

FILE MANIFEST

NO.	DESIGN	STL	SIZE	TRIANGLES	3MF	SIZE
PG-01	Vortex Grip	01_vortex_grip.stl	19.9 MB	398,552	01_vortex_grip.3mf	4.2 MB
PG-02	Cellular Mod	02_cellular_mod.stl	9.9 MB	198,210	02_cellular_mod.3mf	2.1 MB
PG-03	Tessel-Block	03_tessel_block.stl	20.9 MB	417,132	03_tessel_block.3mf	4.5 MB
PG-04	Logic-Grip	04_logic_grip.stl	19.6 MB	391,124	04_logic_grip.3mf	4.1 MB
PG-05	Neuro-Tread	05_neuro_tread.stl	24.2 MB	483,320	05_neuro_tread.3mf	5.1 MB
PG-06	Carbon Matrix	06_carbon_matrix.stl	19.9 MB	397,168	06_carbon_matrix.3mf	4.2 MB
PG-07	Voronoi-Core	07_voronoi_core.stl	11.3 MB	225,794	07_voronoi_core.3mf	2.4 MB
PG-08	Topo-Flow	08_topo_flow.stl	21.7 MB	433,616	08_topo_flow.3mf	4.7 MB
PG-09	Ergo-Contour	09_ergo_contour.stl	4.5 MB	89,324	09_ergo_contour.3mf	0.9 MB
PG-10	Hexa-Mod	10_hexa_mod.stl	17.0 MB	340,206	10_hexa_mod.3mf	3.6 MB

LIMITS OF THIS DOCUMENT

These sheets describe the geometry and its measured properties. They are not a print guarantee: material, printer, cooling and slicer settings all affect the result. Do a preview slice before committing filament. The slot is sized to a ruler measurement of 30 × 16 mm: check your paddle with calipers and adjust PADDLE_W / PADDLE_T in grip_core.py if it differs.